Tacinf: A Unified Tactile Infrastructure for Robot Manipulation

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Abstract

Tactile sensing is essential for robots to achieve force-sensitive manipulation, but current Vision–Language–Action (VLA) models have limited access to aligned vision–tactile–language supervision and lack a common path from language-specified force levels to real-robot force responses. We introduce Tacinf, an integrated tactile infrastructure organized into four complementary components. Tacinf-Tabero replays open-source trajectories through a calibrated tactile simulation stack and turns language-specified force levels into synchronized vision–tactile–language–force data with task and interaction-quality metrics. Tacinf-VTLA instantiates a force-aware vision–tactile–language–action policy on a pretrained VLA, mapping force-related language to fingertip-force targets and closed-loop force–position commands. Tacinf-Tacumi provides a handheld UMI interface with a shared tactile gripper for contact-rich data collection. Tacinf-Taco provides task-aligned Real2Sim calibration, a unified interface for mixed real, simulated, and UMI data, and a policy-level zero-shot Sim2Real evaluation protocol. In the latter protocol, each policy is initialized from a pretrained VLA and its task-specific adaptation uses simulated trajectories only; the adapted policy is then executed on the physical robot under the same force-level instructions without real-robot task demonstrations, policy fine-tuning, or test-time adaptation. Across seven policy variants, the combined task-and-force score reaches a Pearson correlation of $r = 0.847$ between simulation and the real robot. These results position Tacinf as a reusable workflow for language-conditioned force-level following and tactile policy evaluation.

Keywords: Tactile Sensing, Vision-Language-Action Models, Language-Conditioned Force Control, Zero-Shot Sim-to-Real Evaluation

1. Introduction

Physical AI is emerging as a pivotal enabler for robots to operate effectively in the real physical world. For robots to exhibit genuine intelligence in unstructured environments, they must not only perceive their surroundings visually but also comprehend physical laws through direct contact. In humans, touch serves as the most fundamental modality for interacting with objects and is essential for developing physical intuition. While recent advances in vision–language–action (VLA) foundation models have shown remarkable progress Black et al. (2025a,b); Kim et al. (2025a); Liu et al. (2025c); NVIDIA et al. (2025a), these models predominantly rely on internet-scale image–text–video data or robot datasets consisting of image–action pairs collected via specialized hardware. Their reported policy interfaces do not generally include tactile observations, motivating explicit tactile extensions for force-sensitive manipulation. Recent systems collect real tactile data through multimodal sensors and robot-free interfaces Cheng et al. (2026a); Liu et al. (2025a); Wu et al. (2025a); Xu et al. (2025b); Zhao et al. (2025a); nevertheless, scaling synchronized demonstrations across sensors, tasks, and

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embodiments remains challenging. Tactile simulation and unified synthesis platforms provide complementary routes to scale Akinola et al. (2025); Chen et al. (2026); Li et al. (2025c). Our focus is the integration of these capabilities with language-conditioned force control and evaluation.

Building upon VLA models, vision–tactile–language–action (VTLA) models extend perception to include the tactile modality, thereby endowing foundation models with the capacity to interact physically with the world. The central capability is broader than gentle manipulation: a policy should interpret language such as "gently," "softly," or "firmly" as an ordered force-level condition, produce the corresponding contact behavior, and preserve that ordering when executed on a different embodiment. Training such models faces two major challenges. First, VTLA models still require large amounts of aligned vision–tactile manipulation data. Second, existing tactile benchmarks evaluate representation transfer, manipulation success, or force safety under different protocols Chen et al. (2026); Higuera et al. (2025a); Wei et al. (2025); they do not provide a common test of language-conditioned force-level following across simulation and the real robot. Success rates alone do not describe whether a completed manipulation applied the requested force level.

To study language-conditioned force-level following, we introduce Tacinf (Fig. 1), a tactile infrastructure with four complementary components. Tacinf-Tabero replays open-source robot trajectories through a calibrated tactile simulation stack and pairs force-related language with controlled force regimes, synchronized tactile observations, and interaction-quality metrics. Tacinf-VTLA integrates tactile observations into a vision–tactile–language–action architecture and combines force-aware prediction with closed-loop force–position control. Tacinf-Tacumi is a handheld UMI device with a shared tactile gripper for physical data collection. Tacinf-Taco performs task-aligned Real2Sim calibration and reproducible simulation evaluation, and defines a unified interface for mixed real, simulated, and UMI data. Our experiments show that, after initialization from a pretrained VLA, task-specific simulation-only adaptation can preserve force-level ordering on the real robot without real-robot task demonstrations, policy fine-tuning, or test-time adaptation, while retaining high task success.

In summary, our work makes the following contributions:

- **Tacinf-Tabero**, a reusable trajectory-to-tactile data path that converts open-source manipulation trajectories into language-conditioned force-level supervision and synchronized multimodal observations, together with metrics beyond task success.
- **Tacinf-VTLA**, a force-aware VTLA instantiation on a pretrained VLA that maps force-related language to fingertip-force targets and closed-loop force–position commands.
- **Tacinf-Tacumi**, a handheld UMI interface with the same tactile gripper used for deployment, providing a hardware path for collecting contact-rich demonstrations.
- **Tacinf-Taco**, a task-aligned Real2Sim and evaluation path that tests policies after simulation-only task adaptation on the real robot without real-robot task demonstrations, policy fine-tuning, or test-time adaptation, and defines a common interface for future real–simulation–UMI co-training.

The main contribution is the system-level force semantics shared across these components: a language-specified force level is rendered in simulation, predicted by a tactile policy, tracked by the controller, and evaluated on the physical robot. The individual simulator, tactile encoder, haptic interface, and hybrid-control primitives build on the prior work discussed in Section 4; the new claim is their integration into a reusable force-level data and evaluation workflow.

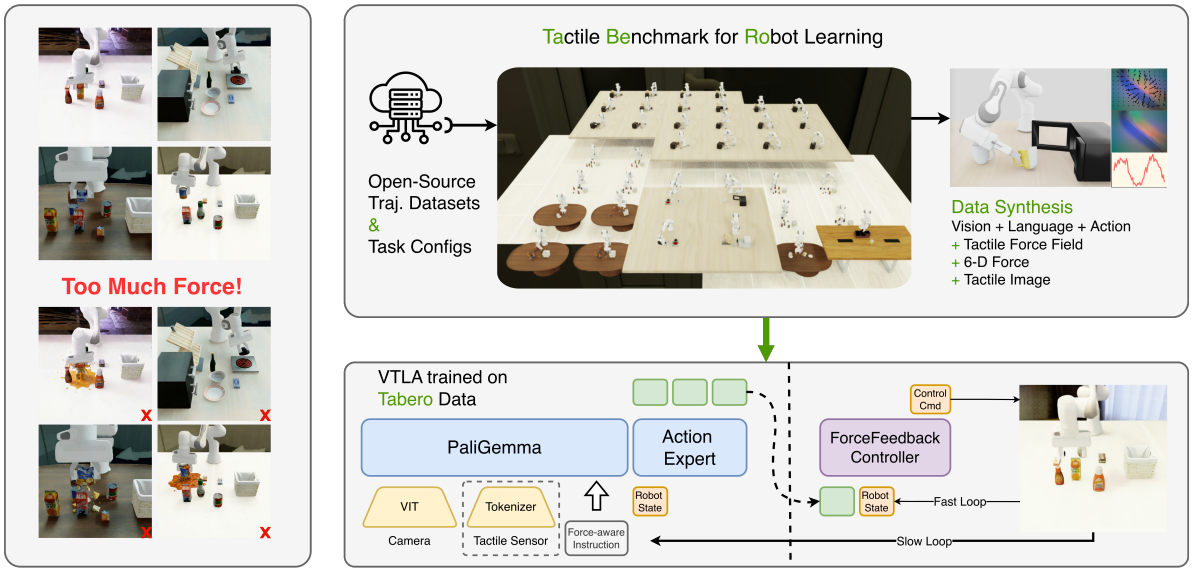


Figure 1: **Overview of the proposed Tacinf framework. Motivation:** Current vision–language–action (VLA) systems and robotic arm–gripper setups based on synthetic data often lack force feedback, causing learned policies to apply unnecessarily large contact forces. **Tacinf-Tabero:** We present a multimodal simulation benchmark integrating Isaac Lab with a calibrated tactile simulation stack and a multidimensional evaluation protocol. The pipeline re-collects open-source trajectories as synchronized streams of multi-view vision, tactile images, force fields, and proprioception. **Tacinf-VTLA:** Leveraging the Tabero data, we instantiate a force-aware tactile policy with a closed-loop force–position controller. Tacinf-Tacumi provides a handheld tactile interface, while Tacinf-Taco performs real-to-sim asset alignment, reproducible simulation evaluation, and mixed-data co-training, and supports subsequent sim-to-real transfer.

2. Tacinf

2.1. Tacinf-Tabero

Tacinf-Tabero is the simulation benchmark component of Tacinf. It provides reproducible language-conditioned tactile manipulation tasks, scalable multimodal data generation, and process-aware metrics for evaluating physical interaction quality.

2.1.1. Data Reuse

Open-source robotic manipulation datasets are valuable community resources. To avoid rebuilding every dataset from scratch, we design a reusable workflow that transcribes datasets originally developed in MuJoCo or other platforms into the Isaac Sim-based Tacinf-Tabero benchmark. The transcription improves visual fidelity through high-quality rendering and augments the data with tactile information.

We first reconstruct task environments in Isaac Lab NVIDIA et al. (2025b) to match the source domains, including assets, initial object and robot poses, and success logic. This workflow can currently be partially automated with SimFoundry Ranawaka et al. (2026). We then replay the source trajectories in the reconstructed scenes to obtain multimodal data. Direct migration of tactile-aware tasks faces two challenges. First, the end-effector differs: our robot uses a gripper integrated with visuo-tactile sensors whose geometry differs from the original. Second, the underlying physical behavior differs: some source datasets rely on Operational Space Control (OSC) Khatib (1987), which is highly sensitive to physical parameters. Datasets collected by real-robot teleoperation or UMI devices also introduce simulation-to-real gaps. We therefore make two adjustments: (1)

align the end-effector tool center point (TCP) by adjusting the robot-arm base pose; and (2) use a high-gain PD joint controller during trajectory replay to minimize cumulative tracking error.

After evaluation, the robotic arm equipped with a tactile gripper achieves an acceptable task success rate during data replay, which we refer to as the data retention rate.

2.1.2. Cross-Modal Data Acquisition

Leveraging the GPU-accelerated parallel rendering capabilities of Isaac Lab, we build a synchronized and scalable multimodal data acquisition system that captures diverse sensory streams with a shared temporal convention. Visual information is obtained from two camera configurations: a wrist-mounted camera and a third-person view camera, both providing synchronized RGB-D data. Tactile observations are produced by a simulated GelSight Yuan et al. (2017) sensor with a resolution of 320×240, providing both RGB tactile images and an 11×9 marker grid. We also implement compatibility with other tactile sensors, such as the Xense sensor, with a resolution of 320×240, providing a 20×11 marker grid. To maximize simulation speed, we implement the tactile perception simulation by following the computation schemes of Taxim and FOTS Si and Yuan (2022); Zhao et al. (2024), which are computationally lightweight.

Tactile Image. Tactile images are generated using the Taxim framework Si and Yuan (2022). GelSight builds on retrographic sensing Johnson and Adelson (2009) and reconstructs contact geometry from calibrated illumination-dependent images Yuan et al. (2017); in simulation, the geometry-to-image mapping is evaluated in the forward direction. In simulation, Taxim moves the height map according to the indentation depth, approximates the gel's static soft-body deformation through multi-layer Gaussian filtering, and then computes surface normals and performs optical rendering. **Marker Displacement Field.**Following the FOTS approach Zhao et al. (2024), we model the displacement field of surface markers on an elastic tactile sensor: given the current object pose and the initial contact state in simulation, we use a Gaussian spatial decay function to simulate the marker displacement caused by normal expansion, shear, and planar torsion. This field provides a raw but informative signal for estimating shear forces and torques. The marker representation used here is specific to optical tactile sensing. Extending the interface to capacitive, piezoelectric, or magnetic sensors would require an independently validated observation model; equivalent marker displacements cannot be assumed to follow from their temporal signal gradients alone.

Contact Force. The physics engine provides the left and right sensor reactions, $\mathbf{f}_L^{S_L}$ and $\mathbf{f}_R^{S_R}$, in their local frames. For real-world measurements, we fit a linear sim-to-real mapping from paired data; its coefficients are released as calibration settings (Appendix D). After rotation into the TCP frame, the negative sum gives the force exerted on the object. The two contact forces and their moment arms provide a point-force approximation of the TCP wrench (Section 2.2); the scalar grip proxy is computed separately from the local normal components.

All cameras are rendered in parallel using tiled rendering, and all modalities, including visual, tactile, force, language instructions, and executed actions, are sampled synchronously at 20 Hz to produce temporally aligned multimodal observations at each time step.

2.1.3. Language-Conditioned Force-Level Data Generation

We focus on tasks in which the desired interaction cannot be specified adequately by task completion alone, including explicit force instructions and contact-sensitive object handling. We distinguish two categories: (1) language-conditioned force-level tasks; and (2) tasks oriented toward deformable objects or objects with articulation parts. Tabero is primarily used to validate the first category, while SoftVTBench Jing et al. (2026) evaluates deformable-object manipulation using independent deformation ground truth and deformation-aware success criteria. Limited by the accuracy ceiling of the simulator, we do not build our simulation benchmark on the second category; instead, we validate the corresponding behavior directly in real-world scenarios.



Figure 2: **Tacinf-Tabero simulation benchmark.** Tacinf-Tabero replicates the LIBERO task environments, enables data reuse, enhances the visual fidelity of simulated data, and makes it possible to obtain high-quality tactile modalities.

For the first category of tasks, most open-source datasets use continuous actions for arm control but binary or discrete gripper commands, limiting fine-grained force regulation and producing narrow within-task force distributions. We address this by executing the same task with different force magnitudes through low-level gripper-controller parameters.

In our simulation, the gripper uses a spring–damper position servo, consistent with the impedance-control formulation Hogan (1985). For a single-DOF gripper with width p, the commanded force is:

$$F^{\text{cmd}} = K_p(p^{\text{target}} - p) + K_d(\dot{p}^{\text{target}} - \dot{p}). \tag{1}$$

At steady state after contact, velocities vanish and the static grip force approximates $F_{\text{static}} \approx K_p \cdot \delta$, where $\delta = p - p^{\text{target}} > 0$ is the penetration depth. Thus, varying K_p directly scales the steady-state force under identical high-level commands. During initial contact, however, the impact force is dominated by damping: $F_{\text{impact}} \propto K_d \cdot |\dot{p}|$, so adjusting K_d diversifies transient forces. This variation is difficult to realize directly with many real robot grippers, motivating the tactile UMI device described below for data collection.

We replay trajectories under different (K_p, K_d) settings to instantiate an ordered force-level condition ℓ. In the reported benchmark, $\ell \in \{10\%, 25\%, 100\%\}$, with the same task and trajectory semantics retained across levels. These interactions produce distinct sensor-side force patterns $\mathbf{f}_L^{S_L}$ and $\mathbf{f}_R^{S_R}$; their TCP-frame transforms and object-side signs follow Eq. (3). Following Tactile-VLA Huang et al. (2025c), we attach language such as "gently" or "softly" to low-force conditions and "firmly" or "tightly" to high-force conditions, aligning the language condition with the measured force distribution. Here "gentle" is one endpoint of a force-level following problem rather than the complete objective: the policy should preserve the ordering and separation induced by ℓ. For special tasks whose nature already demands substantial force, such as wiping a blackboard, we do not add an adverb. We also record continuous gripper aperture p rather than discrete open/close signals, preserving compatibility between the action space and force dynamics. This pipeline provides a reusable way to generate language-conditioned force profiles for arbitrary tasks.

2.1.4. Metrics Beyond Success Rate

Task success alone overlooks critical aspects of physical interaction; force-aware and deformation-aware benchmarks address complementary aspects of this issue Jing et al. (2026); Wei et al. (2025). Because the current version does not include deformable objects, we adopt a rigid-body plus force-sensing approach to approximate the contact process. We introduce the forces obtained at the gripper fingertips as evaluation metrics to quantify the quality of physical interaction during task execution: **Maximum Transient Grip Force**

(MG). The average of the top 5% grip force values over an episode, capturing peak grasping effort and transient spikes that may indicate aggressive or damaging behavior. **Average Grip Force (AG).** The mean grip force during contact, reflecting the nominal force used in stable grasping. **Maximum Transient Applied Force (MA).** The average of the top 5% magnitudes of the applied force, characterizing extreme interaction events such as impacts. **Average Applied Force (AA).** The mean applied force magnitude during contact, measuring overall interaction intensity.

With wrench-aware control, the three translational forces and the two modeled torque components about TCP x and z are additionally tracked in terms of their means, peaks, and tracking errors. Under the sufficient-grip, no-slip assumption below, object-dependent torque about the grasp axis is excluded from this sensor-centered evaluation. Together, these metrics enable a more nuanced and physically grounded evaluation of robot policies, moving beyond binary success to assess safety and interaction quality.

2.2. Tacinf-VTLA

Tacinf-VTLA is not a specific model architecture but a methodology that equips foundation vision–language–action models with tactile sensing and language-conditioned force-level control; this paper instantiates it on the OpenPi infrastructure associated with π_0 Black et al. (2025b) as a concrete example. The instantiated policy component predicts fingertip-force targets from vision, language, and touch and incorporates a closed-loop force–position control strategy that converts its outputs into compliant robot motion.

2.2.1. Fingertip-force Decomposition and Two-contact Wrench Mapping

The separation of internal grasp forces from the resultant object wrench is standard in grasp mechanics Lynch and Park (2017); recent compliant imitation-learning systems also distinguish grasp and external interaction forces Choi et al. (2026). In our implementation, a two-finger parallel gripper has a single opening–closing degree of freedom, so the planar contact at its fingertips decomposes naturally into two parts: the components squeezing each other along the closing axis form the *grip force*, while the remaining components constitute the *applied interaction* on the object, represented as an estimated six-dimensional wrench at the TCP under a two-contact approximation (3D force plus a 3D moment vector whose grasp-axis component is neglected). During grasping, the normal force that compresses the sensor pads against the object surface dominates; when the gripper transports, wipes, or twists the object, shear forces and moments equally determine the object motion.

2.2.2. Hybrid Controller

Building on hybrid position–force control and impedance-based interaction control Hogan (1985); Raibert and Craig (1981), as well as their use in tactile-conditioned policies Choi et al. (2026); Huang et al. (2025c), we implement force-error-driven pose and gripper-width corrections. We distinguish a scalar grip channel from the applied wrench, which comprises a 3D force and a 3D torque. Let $i \in \{L, R\}$ index the left and right fingertips, S_i denote each sensor frame, and T denote the TCP frame. A superscript S_i or T specifies the coordinate frame, whereas $\top$ denotes transpose. Lowercase $\mathbf{f}_i^{S_i} \in \mathbb{R}^3$ is the contact reaction acting on sensor i; its local z-axis is normal to the pad. The grip channel uses the conservative symmetric-contact proxy

$$F_{\text{grip}} = \min\left(|f_{L,z}^{S_L}|,\ |f_{R,z}^{S_R}|\right) \in \mathbb{R}_{\geq 0}. \tag{2}$$

Here F_{grip} is a one-dimensional force magnitude (N) computed from sensor-local normal components, not a three-dimensional force vector in the sensor frame. For symmetric squeezing, the two normal forces acting on the object are equal and opposite, so the squeeze magnitude is the force at one contact rather than their sum. Taking the smaller magnitude keeps the proxy conservative for asymmetric contact; any unbalanced normal force also contributes to the resultant applied force.

The TCP frame T has x along approach, y along the grasp axis, and z completing the right-handed triad. Lowercase $\mathbf{f}_L^T, \mathbf{f}_R^T$ are the two sensor reactions expressed in this common frame; the subscripts identify the contacts, not separate TCP resultants. Capital $\mathbf{F}^T$ denotes the single resultant force exerted on the object. Rotation and action–reaction give

$$\begin{aligned} \mathbf{f}_L^T &= \mathbf{R}_{S_L}^T \mathbf{f}_L^{S_L}, \qquad \mathbf{f}_R^T = \mathbf{R}_{S_R}^T \mathbf{f}_R^{S_R}, \\ \mathbf{F}^T &= -(\mathbf{f}_L^T + \mathbf{f}_R^T). \end{aligned} \tag{3}$$

The sensor-local normal axes map to opposite directions along TCP y; their magnitudes enter Eq. (2). The scalar aperture p is the distance between the two contact planes along y, distinct from the bold TCP position vector $\mathbf{p}$. Approximating the contacts by the pad centers, at $(0, -p/2, 0)^\top$ and $(0, p/2, 0)^\top$ relative to the TCP, gives the moment directly from the two forces Lynch and Park (2017):

$$\boldsymbol{\tau}^T \approx \frac{p}{2}\big(f_{L,z}^T - f_{R,z}^T,\ 0,\ f_{R,x}^T - f_{L,x}^T\big)^\top. \tag{4}$$

We assume the grip force is sufficient to maintain sticking contact, with no relative slip between the object and the tactile pads. Under this assumption, torque about the grasp axis y depends on the object and its contact conditions. To keep the calculation centered on the sensors and known gripper geometry, we leave this object-dependent component outside the modeled wrench and its feedback correction. The y-moment entry is therefore set to zero as an omission convention. Sticking contact can transmit a nonzero torque; the zero entry does not assert that the physical torque vanishes.

The resulting single TCP wrench combines the resultant force (N) and moment (N·m):

$$\mathbf{W}^T = \begin{bmatrix} \mathbf{F}^T \\ \boldsymbol{\tau}^T \end{bmatrix} \approx \begin{bmatrix} -(f_{L,x}^T + f_{R,x}^T) \\ -(f_{L,y}^T + f_{R,y}^T) \\ -(f_{L,z}^T + f_{R,z}^T) \\ \frac{p}{2}(f_{L,z}^T - f_{R,z}^T) \\ 0 \\ \frac{p}{2}(f_{R,x}^T - f_{L,x}^T) \end{bmatrix}. \tag{5}$$

The grip proxy is one scalar computed from sensor-local normals, whereas $\mathbf{W}^T$ contains three force and three moment entries in the TCP frame. Five wrench entries are computed from the two-point-force mapping; the grasp-axis entry is omitted under the stated modeling scope. Symmetric squeeze cancels from the resultant wrench and is regulated separately through the gripper aperture.

Wrench-Aware Control The policy predicts an end-effector pose $\mathbf{P}^{\text{pred}} \in SE(3)$ and target sensor reactions in the two local frames. Each target and measurement is rotated into the TCP frame before subtraction:

$$\begin{aligned} \mathbf{e}_L^T &= \mathbf{R}_{S_L}^T(\mathbf{f}_L^{S_L,\text{target}} - \mathbf{f}_L^{S_L}), \\ \mathbf{e}_R^T &= \mathbf{R}_{S_R}^T(\mathbf{f}_R^{S_R,\text{target}} - \mathbf{f}_R^{S_R}). \end{aligned} \tag{6}$$

The same sum and difference used in Eq. (5) give the force-driven pose corrections directly:

$$\begin{aligned} \Delta\mathbf{p} &= -\mathbf{K}_p \mathbf{R}_T^B(\mathbf{e}_L^T + \mathbf{e}_R^T), \\ \Delta\boldsymbol{\phi} &= \mathbf{K}_\phi \mathbf{R}_T^B \begin{bmatrix} \frac{p}{2}(e_{L,z}^T - e_{R,z}^T) \\ 0 \\ -\frac{p}{2}(e_{L,x}^T - e_{R,x}^T) \end{bmatrix}, \\ \mathbf{p}^{\text{cmd}} &= \mathbf{p}^{\text{pred}} + \Delta\mathbf{p}, \qquad \mathbf{R}^{\text{cmd}} = \exp\big([\Delta\boldsymbol{\phi}]_\times\big)\mathbf{R}^{\text{pred}}. \end{aligned} \tag{7}$$

Here $\mathbf{R}_T^B$ rotates from TCP to base coordinates while moments remain referenced to the TCP. The gains $\mathbf{K}_p$ and $\mathbf{K}_\phi$ have units $\mathrm{m/N}$ and $\mathrm{rad/(N{\cdot}m)}$, respectively, and $[\,\cdot\,]_\times$ denotes the cross-product matrix. Consistent

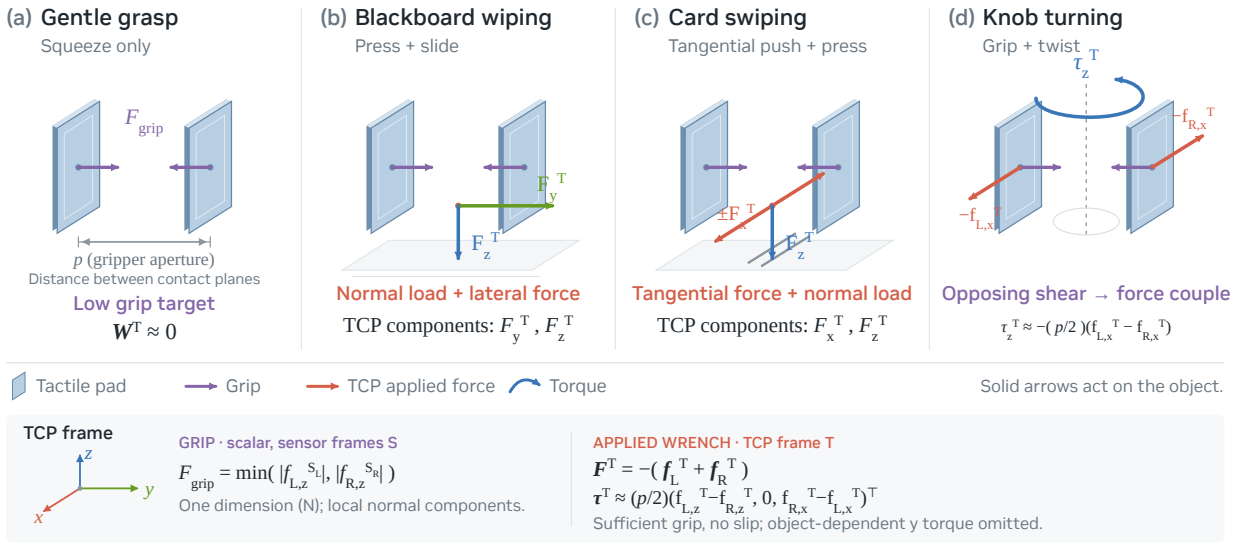


Figure 3: **Fingertip-force decomposition across tactile skills.** Purple straight arrows show internal grip; TCP directions use x red, y green, and z blue throughout. The aperture p is the distance between the two tactile contact planes along y. (a) Gentle grasping isolates the scalar grip channel in an idealized unloaded case; the object silhouette is omitted. (b) Blackboard wiping combines lateral F_y^T and downward F_z^T. (c) Card swiping combines bidirectional F_x^T and downward F_z^T. (d) Opposing contact forces $-f_{L,x}^T$ and $-f_{R,x}^T$ generate τ_z^T. Capital $\mathbf{F}^T$ is the single resultant force on the object; lowercase $\mathbf{f}_L^T, \mathbf{f}_R^T$ are sensor reactions rotated into the TCP frame. Negative signs convert reactions to forces on the object. The scalar grip proxy uses sensor-local normal components (Eq. (2)); the TCP wrench uses the two-point-force approximation in Eq. (5). Assuming sufficient grip to prevent relative slip, the sensor-centered model omits the object-dependent y-axis torque; its zero entry denotes omission, not zero physical torque. Contact planes and arrows are schematic and not to scale.

with the sufficient-grip, no-slip assumption and sensor-centered scope, the controller omits object-dependent grasp-axis torque feedback. The zero middle entry adds no correction in that channel before the base-frame rotation and gains. The policy's predicted orientation still specifies rotation about all axes, but this does not provide grasp-axis torque feedback. The rotation vector $\Delta\phi$ updates orientation through the exponential map. Differential inverse kinematics Lynch and Park (2017) converts the corrected pose into joint velocity commands, while the independent grip loop adjusts aperture.

Grip Force Control To overcome the limitation of position control in precise force regulation, we implement a grip force feedback loop. The predicted sensor-frame reactions $\mathbf{f}_i^{S_i,\text{target}}$ provide their local normal components directly, giving $F_{\text{grip}}^{\text{pred}}$ through Eq. (2); the grip-loop target is

$$F_{\text{grip}}^{\text{target}} = (1 + k_{\text{ff}})\, F_{\text{grip}}^{\text{pred}}, \quad k_{\text{ff}} > 0, \tag{8}$$

where k_{ff} is a feedforward gain. The measured grip force $\tilde{F}_{\text{grip}}$ is exponentially smoothed. Let p denote the gripper width. We apply a correction Δp only when $|\Delta p| \geq \text{dz}$, where $\text{dz} > 0$ is a deadzone threshold accounting for gripper imprecision:

$$\Delta p = k_p^{\text{adm}} \cdot (F_{\text{grip}}^{\text{target}} - \tilde{F}_{\text{grip}}), \tag{9}$$

with k_p^{adm} an admittance gain. The final gripper command is

$$p^{\text{cmd}} = \text{clip}\big(p^{\text{pred}} + \Delta p,\ 0,\ p_{\max}\big), \tag{10}$$

where p^{pred} is the policy-predicted gripper width. This allows the policy to specify a desired grip force while the controller achieves accurate tracking via feedforward and force feedback. In summary, the two fingertip

force triples together with the aperture provide the signals used by the low-level controller: a grip-force proxy from the normals, and estimated wrench-related pose corrections from the sum and the half-aperture-scaled difference, while task-level coordination is delegated to the policy.

2.2.3. Tactile Policy

Tacinf-VTLA is trained on Tacinf-Tabero data and consumes the tactile representations and force-aware instructions produced by the infrastructure. To integrate tactile signals into the VLA foundation model, we evaluate three tactile input adapters that map different tactile modalities into conditional tokens. The high-level policy then jointly predicts future gripper poses and six-dimensional fingertip-force setpoints based on vision, language, and touch. These commands are tracked by the fixed admittance-based low-level controller described above (Fig. 4), which provides physical compliance; the high-level policy learns to reason about appropriate interaction forces given the task context. Building on the OpenPi infrastructure Black et al. (2025b) and leveraging flow matching Lipman et al. (2023), our approach enables continuous prediction of both pose and the six fingertip force components. The three adapters share the same temporal window, preprocessing, backbone, and training protocol; only the tactile input interface is varied in the ablation. Below, we detail the adapters and loss function, and compare alternative tactile injection strategies inspired by prior work.

Force-Field Adapter. It processes tactile marker displacement fields relative to the initial rest state of the sensor. The input consists of $H+1$ frames: the first frame captures the undeformed marker layout, and the subsequent H frames record 2D marker positions in the sensor's local plane, resulting in an array $M \in \mathbb{R}^{(H+1)\times N\times 2}$. Here, N denotes the number of markers, and the 2D element denotes the marker's planar coordinates. A lightweight Temporal Convolutional Network (TCN) encodes this spatiotemporal sequence into tokens for integration into the transformer backbone. Detailed architecture and hyperparameters are provided in Appendix A.

Tactile-Image Adapter. Inspired by visual–tactile policy conditioning Xue et al. (2025); Zhang et al. (2026), we implement an image adapter for our backbone. Specifically, we construct a single composite tactile image by arranging the H historical frames from the left finger and the H frames from the right finger into a unified spatial layout. The value of H is determined by the shared multi-view image resolution and the prescribed tactile-image resolution; when $H = 1$, the composite contains only the current contact state. This image, encoding the full tactile history of both fingertips, is processed by the same visual encoder used for RGB-D observations. Its features then interact with visual features via cross-attention in the transformer, enabling joint reasoning over contact and scene geometry. Implementation details and parameter settings are provided in the Appendix A.

Force Adapter. We use the 3D sensor-frame reaction vectors $\mathbf{f}_L^{S_L}$ and $\mathbf{f}_R^{S_R}$ as raw inputs, resulting in a 6-dimensional force signal. Although these reactions can be transformed and combined with the known fingertip geometry to compose the applied wrench at the TCP, we find it more effective to directly feed their concatenation into a multilayer perceptron (MLP) to obtain a compact latent representation of contact dynamics. This representation is injected into the model alongside other modalities to enhance physical awareness. Specific network configurations and training parameters are provided in the Appendix A.

Force Supervision. Flow matching Black et al. (2025b); Lipman et al. (2023) supports continuous-valued outputs, including force components, motivating our weighted loss design. The action vector a includes both pose commands and six sensor-frame force components, $[\,\mathbf{f}_L^{S_L}\ ;\ \mathbf{f}_R^{S_R}\,]$. For a batch b and time $t \sim \mathcal{U}(0,1)$, we define the interpolated input $x_t = (1-t)\epsilon + ta$ and target velocity $u_t = a - \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. The prediction error $e = v_t - u_t$ is split into action and force components, $[e^{(\text{act})}, e^{(\text{force})}]$. We normalize the six force dimensions to a common scale and upweight them by a factor $\lambda_{\text{force}} > 0$, while keeping other dimensions at

unit weight. The final loss is:

$$\mathcal{L} = \frac{1}{D_{\text{act}}} \sum_{d=1}^{D_{\text{act}}} w_d (e_d)^2, \tag{11}$$

where $w_d = \lambda_{\text{force}}$ for dimensions corresponding to the predicted fingertip forces, and $w_d = 1$ otherwise, with $\lambda_{\text{force}} > 0$. Although the six components share the same unit (N), their magnitudes differ across directions, so the normalization is necessary for stable six-dimensional supervision.

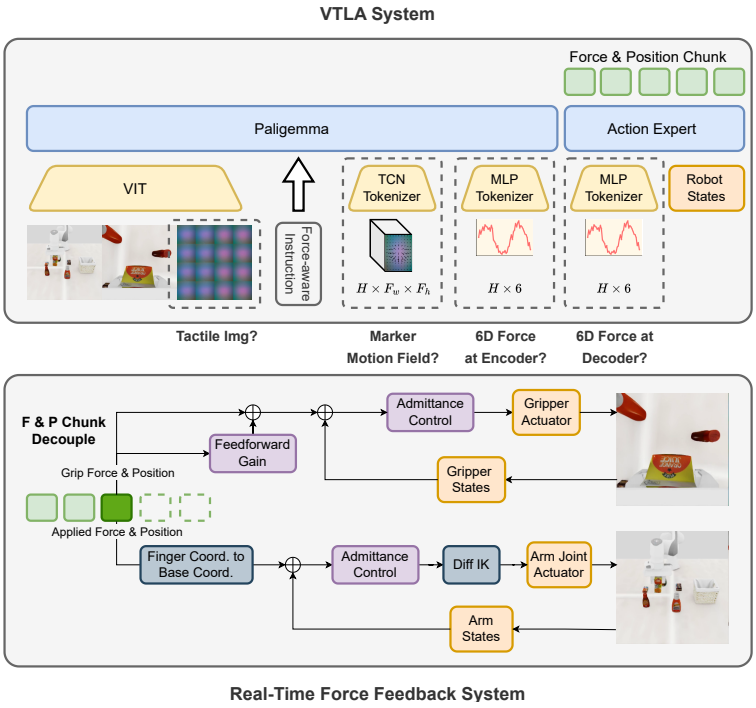


Figure 4: **Tacinf-VTLA system overview.** Tactile inputs are encoded by input adapters and fused with vision and language. The policy predicts force–position commands, which its two-loop closed-loop controller tracks to achieve compliant interaction.

2.3. Tacinf-Tacumi

To support end-to-end learning of fine-grained force, touch, language, and action, we design and develop Tacinf-Tacumi, a multimodal universal manipulation interface that follows handheld robot teaching Chi et al. (2024) and tactile demonstration collection Cheng et al. (2026a); Liu et al. (2025a); Wu et al. (2025a); Xu et al. (2025b). Our implementation combines motor-torque feedback with a shared collection and deployment gripper. It provides a compact interface for collecting synchronized tactile observations and device motion during real-world demonstrations. A common data schema associates these signals with scene observations and task instructions, allowing Tacumi demonstrations to complement Tacinf-Tabero trajectories and directly support Tacinf-VTLA training. Quantitative evaluation in this paper focuses on the Tabero simulation benchmark; Tacumi provides the hardware path for extending the same protocol to real experiments and for supplementing VTLA data.

The handheld device follows a master–slave design with a shared gripper/sensor assembly, low manufacturing cost, and straightforward reproducibility. Its three functional parts are: (1)Master: Held and operated by a human. A micro servo motor and its lever encode the operator's gripper open/close intent; the same servo acts as the haptic actuator for torque feedback. (2)Slave: A parallel gripper that performs the physical grasp,

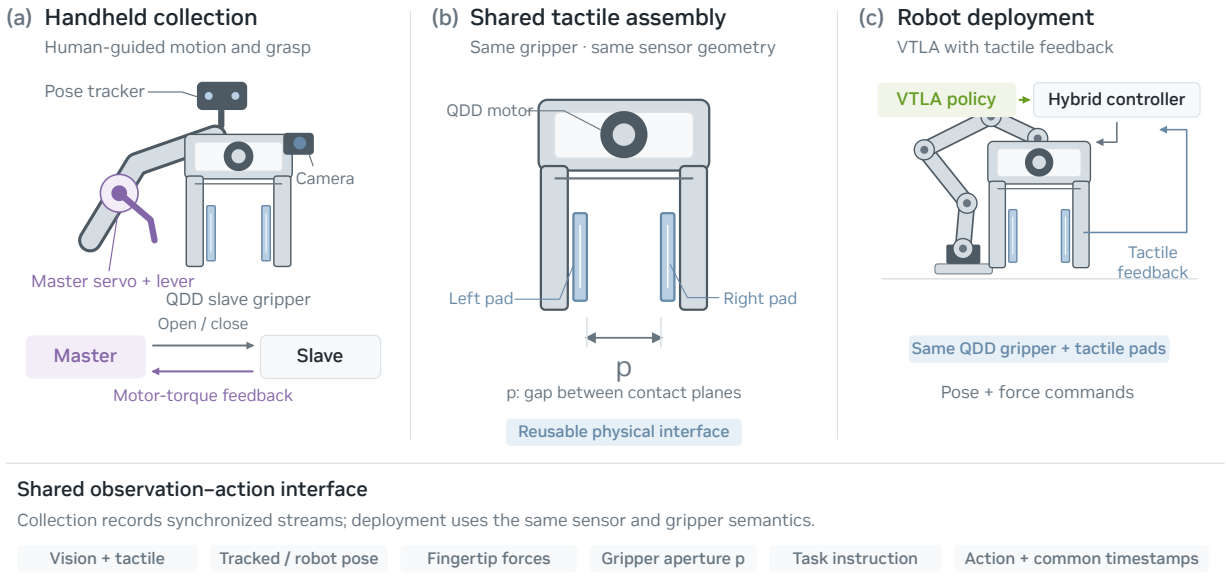


Figure 5: **Tacinf-Tacumi: a shared tactile interface for collection and deployment.** (a) The handheld master commands the QDD slave gripper and receives motor-torque feedback; a tracker, camera, and tactile sensors supply synchronized observations. (b) Collection and robot deployment reuse the same parallel gripper and two tactile pads. The aperture p is the gap between the contact planes. (c) VTLA commands the robot through the hybrid controller with tactile feedback. The common observation–action interface connects collected trajectories to policy inputs and targets. Motor-torque feedback is distinct from the estimated fingertip contact forces. Hardware illustrations are schematic and not to scale.

driven by a high-dynamic-response quasi-direct-drive (QDD) motor with high current–torque transparency. (3)Perception: High-resolution Xense vision-based tactile sensors are integrated at the gripper fingertips. Their elastomer marker arrays and optical cameras capture high-frequency microscopic 3D deformation of the contact surface and output tactile images and contact-force-field features during operation.

Figure 5 summarizes the shared hardware and the separate command, haptic-feedback, and observation paths.

2.3.1. Torque-Based Closed-Loop Hybrid Control

We implement a motor-torque-based closed-loop force–position controller that provides an operator haptic feedback channel during data collection, following established position/force-feedback approaches Hogan (1985); Raibert and Craig (1981). The system operates in two stages: (1)Free-motion stage (position control): Before contact, the system operates in position-control mode. The master servo encoder angle is read at high frequency and mapped to a target position for the slave QDD motor, driving the parallel gripper synchronously. (2)Physical-interaction stage (closed-loop force feedback): When the tactile sensor detects contact or the gripper approaches an object, the system directly reads the slave QDD motor's streaming torque τ_{slave}. Its current–torque transparency avoids the need for an expensive external six-axis force/torque sensor.

The torque is mapped through a feedback gain and synchronously converted into a reverse torque-current command for the handheld master servo:

$$\tau_{\text{master}} = k \cdot \tau_{\text{slave}} \tag{12}$$

where k is the feedback gain. This loop renders slave-motor torque at the handheld actuator, providing a bidirectional haptic channel intended to help the operator modulate grasp effort. Unlike a calibrated

fingertip wrench measurement, motor torque also contains contributions from transmission friction and actuator dynamics; therefore, this feedback is not by itself a measurement of object stiffness or slip.

2.3.2. Tracked Hand-Pose Acquisition

To record the operator's 6-DoF hand trajectory in 3D space, we use a lightweight consumer VR tracker rather than an expensive external optical motion-capture system such as OptiTrack. By combining inside-out visual-inertial SLAM with infrared tracking, the tracker provides a tracked 6-DoF hand pose under changing illumination and background conditions; the achieved accuracy depends on the tracker and scene conditions.

2.3.3. Shared Gripper/Sensor Assembly for Deployment

Prior interfaces address collection-to-deployment differences through hardware design, calibration, or adaptation Chi et al. (2024); Xu et al. (2025a); Yu et al. (2025a). Remaining differences can arise when the collection handle and deployment gripper have different geometry or dynamics. Tacinf-Tacumi instead uses the same gripper and tactile-sensor assembly on the UMI side and the robot deployment side. In deployment, the robot arm directly mounts the same QDD parallel gripper and Xense tactile sensors used by the handheld device. This shared assembly provides two benefits: (1) Low-cost transfer. The deployment system reuses the low-cost QDD gripper and avoids dependence on commercial industrial grippers costing thousands of dollars. (2) Reduced gripper-side mismatch. The gripper appearance, optical occlusion, fingertip shear response, and kinematic parameters (such as maximum stroke and grasp stiffness) are identical during data collection and robot execution. This reduces gripper-specific visual and structural differences while leaving arm, camera, and operator-to-policy differences to be aligned separately. Arm dynamics, camera placement, and operator-to-policy differences still require alignment; shared gripper morphology does not by itself establish out-of-distribution generalization.

2.4. Tacinf-Taco

Tacinf-Taco (Tactile policy Co-training) connects real, simulated, and UMI data for joint VTLA post-training. Real trajectories provide the task and interaction seed; task-aligned simulation expands object and background variation and uses Tacinf-Tabero replay to generate synchronized tactile data; Tacinf-Tacumi adds demonstrations with varied object positions, orientations, and approach motions. These complementary sources are mixed for joint post-training of one VTLA policy, whose physical deployment provides the next collection context. Figure 6 summarizes this system design.

The framework consists of three dependent parts:

1. **Task-aligned scene reconstruction:** transfer workspace appearance and interactive-object geometry into a common task frame, providing the assets for controlled visual variation.
2. **Manipulator-system Real2Sim:** align the robot, cameras, tactile sensing, and action interfaces so that replay produces synchronized observations and force targets compatible with real data.
3. **Augmentation and post-training:** combine simulation-based visual and force-level variation with UMI-based spatial variation, then return the resulting VTLA policy to the shared physical interface.

The aligned simulator also supports policy evaluation. The simulation-only adaptation protocol and paired Sim/Sim2Real evaluation are specified in Sections 3.5 and 3.6; they are separate from the mixed-data post-training design described here.

The reusable unit is the interface contract rather than a particular asset or backbone. A new source dataset is mapped to the task/state/success schema, a new tactile sensor supplies a calibrated observation model, and a new pretrained VLA supplies its own input adapter and action head while retaining the common timestep, normalization, force-supervision, and evaluation interfaces. This makes the framework extensible by design,

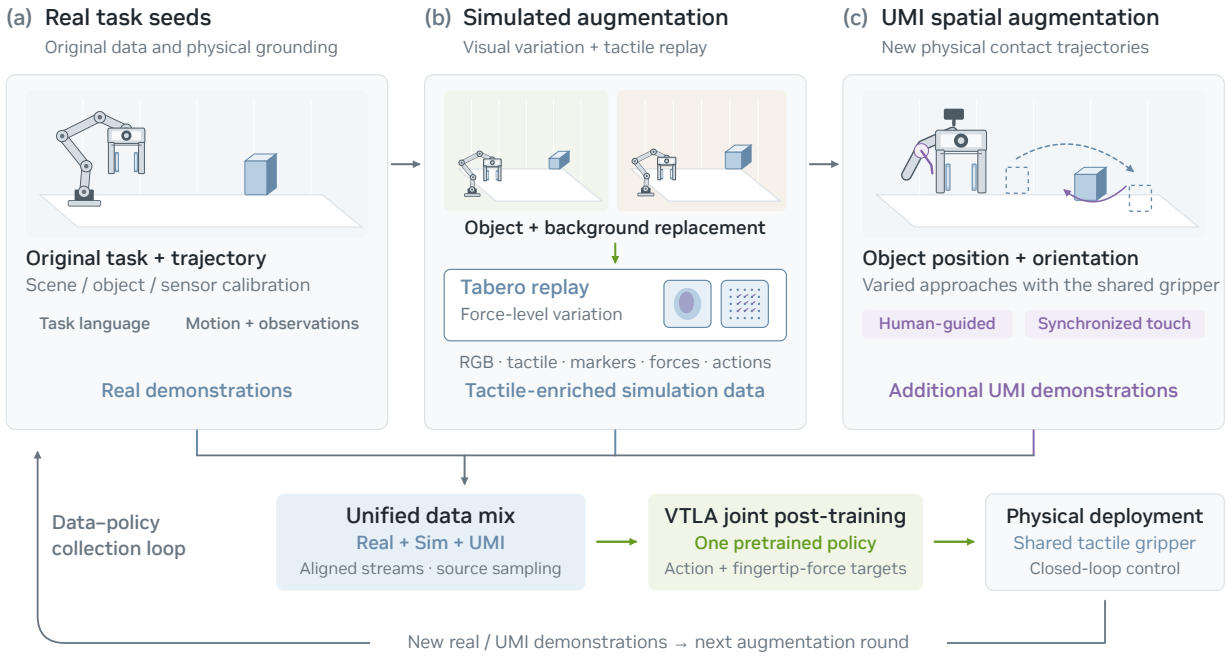


Figure 6: **Tacinf-Taco: the augmentation and joint post-training loop.** (a) Real trajectories seed the task and anchor scene and sensor calibration. (b) Simulation varies objects and backgrounds; Tabero replay generates synchronized tactile observations, force labels, and actions under controlled force levels. (c) Tacumi collects additional physical demonstrations with varied object positions, orientations, and approaches. These task-expansion stages contribute separate real, simulated, and UMI trajectories to a common pool for joint VTLA post-training. Deployment on the shared tactile hardware informs the next collection round, closing the data loop across training iterations. Scene and tactile illustrations are schematic, not experimental observations. This figure describes the co-training design; its empirical evaluation is specified separately in Section 3.8.

while quantitative generalization to additional datasets, sensors, and embodiments remains an empirical question.

Following real-to-sim evaluation approaches that align policy-relevant observations and control Jain et al. (2026); Li et al. (2025b), and tactile sim-to-real studies such as AnyRotate and SimShear Freud et al. (2025); Yang et al. (2025), Taco uses two practical criteria. First, **visual–physical separation**: Gaussian or neural appearance models are used only for rendering and never for contact; explicit collision geometry handles contact, support, and shadow reception. Second, **policy-relevant consistency rather than absolute-parameter equality**: parameters are adjusted to match the data structure, distribution, and visual effect that matter to the policy, because real physical parameters often require time-consuming and extensive calibration and measurement, and some are unavailable under commercial restrictions. In practice, we check contact onset, contact-patch persistence, force scale, marker response, and action replay. These checks distinguish our protocol from a claim of pixel-level digital-twin fidelity.

2.4.1. Task-Relevant Scene Reconstruction

This module transfers the real workspace and interactive objects into Isaac Sim, enabling rendering and interaction under the same camera extrinsics and operation region. The workspace is represented visually by 3D Gaussians, while objects enter the physical layer as independent textured meshes. Both are aligned to the task coordinate system and obey the visual–physical separation principle.

Task-Aligned Workspace Reconstruction. We represent workspace appearance with 3D Gaussian Splatting (3DGS) Kerbl et al. (2023) and convert it into an Isaac Sim-loadable NuRec Volume. The task-side robot, white collision table, cameras, and operation region remain unchanged; the Gaussian asset is aligned to the task frame using a single registration matrix $\mathbf{T}_{\mathrm{reg}}$. A spatial camera continuously records trajectories around the operation region that cover the reachable space and fields of view of the third-person and wrist cameras, while logging calibrated images, poses, and COLMAP intrinsics/extrinsics Schonberger and Frahm (2016). The reconstruction is exported as a standard 3DGS point set in a Z-up, meter-based convention. The capture focuses on the camera frusta used by the policy rather than uniformly covering the entire room.

3DGS Post-Processing. During scanning, the robot arm is present in the scene, so retaining its Gaussians would create double geometry with the simulated URDF robot. Under a tabletop-plane constraint, we perform RANSAC plane estimation Fischler and Bolles (1981), candidate-point selection with tabletop protection, connected-component clustering and morphological filtering, and multiple cleanup passes based on bounding-box neighborhoods, nearest-neighbor growth, and base-column constraints, while protecting fine structures such as cables wrapped around the arm. For rendering compatibility, we remove far-field outliers, contract higher-order spherical harmonics to DC-only, and ensure that the declared SH order matches the specular tensor dimensions, zero-padding missing dimensions when necessary. Finally, the scene USD uses an outer transform to cancel the task's fixed external frame and retains only $\mathbf{T}_{\mathrm{reg}}$ in the inner transform relative to the robot base.

Object Reconstruction. Interactive objects enter the task as independent USD assets rather than being extracted from workspace Gaussians. For each real object, we capture one image and use Seed3D 2.0 Gu et al. (2026) to generate a textured 3D model exported as USD. The explicit mesh follows Isaac's rasterization path; when Gaussian appearance and an object mesh overlap, the object asset owns the interaction while the Gaussian layer supplies only unmodeled background. The exported USD has no collision geometry. We therefore add rigid-body properties through a USD overlay and generate collision meshes with convex decomposition without modifying the generated asset itself. The visual mesh is used only for rendering and the collision mesh only for contact; object physics is supplied by the overlay. Because Seed3D scale and initial pose can differ from the real object, we import the asset into Isaac Sim, replay the recorded real demonstrations, and iteratively adjust object scale and initial position along each axis until grasp and placement trajectories align. The calibration objective is to improve trajectory and success-rate consistency between simulated replay and the real robot, rather than to minimize appearance error alone; during this calibration stage, task demonstrations and robot trajectories remain unchanged. Appendix E lists the scene-verification checks applied to the reconstructed assets before evaluation.

2.4.2. Embodiment Real2Sim

Scene alignment alone is insufficient for policy transfer: the simulated embodiment must expose the same action and observation semantics as the real system. To ensure the generality of our framework, we do not reconstruct the robot arm's complete Gaussian appearance; instead, we reproduce only its dynamics and kinematics system. In addition, we explicitly simulate the shared Tacumi gripper, including its gripper control and tactile sensing.

Manipulator System. We build the XArm dynamics in Isaac Lab and tune its PD parameters to match the real robot's control behavior. Hand–eye calibration provides the third-person camera pose relative to the robot base. Because the Tacumi gripper uses the shared gripper/sensor assembly, the wrist-camera relationship is known from its mechanical construction rather than estimated from an external sensor model. Finally, we tune the shared tactile gripper's passive-joint mode, PD parameters, and the force–position hybrid controller introduced above.

Tactile Observation Modeling. Our tactile Real2Sim path combines the Taxim optical model Si and Yuan (2022)

and the FOTS marker-motion model Zhao et al. (2024) in a shared Isaac Lab sensor interface. The simulator first derives a contact representation from the depth image beneath the pad. Taxim maps local surface geometry to a tactile RGB observation, while FOTS maps contact deformation and relative tangential/rotational motion to marker flow. The two models are therefore complementary forward models: they reproduce the signals exposed to the policy without requiring an exact replica of the commercial sensor's internal optics or elastomer mechanics. Following the calibration methods of Taxim and FOTS, we calibrate the above models using a small set of real tactile observations and, through parameter settings in Isaac Sim, align the simulated and real data streams in resolution, channel semantics, preprocessing, and temporal-update conventions. We jointly adjust the observation mapping and the contact geometry. In particular, the normal thickness of the tactile-pad collision volume is treated as an asset-level variable and tuned to reduce discrete-time interpenetration while preserving a stable contact patch and consistent activation of both tactile models. This calibration targets policy-relevant contact responses rather than visual mesh agreement alone. The appendix summarizes the calibration procedure and replay-based acceptance checks; the complete simulator configuration is released with the calibrated asset and pinned by the run manifest. Due to commercial restrictions, we cannot obtain the complete optical and mechanical specifications of Xense. We therefore only tune the cameras' rendered view and operating depth range at the observation level, and do not claim strict structural extrinsic registration.

2.4.3. Data Augmentation and Joint Post-Training

Taco combines three data sources to close the system loop: $\mathcal{D}_{\text{real}}$, $\mathcal{D}_{\text{sim}}$, and $\mathcal{D}_{\text{umi}}$. The goal is to expand the coverage of an initial real task while preserving consistent language, contact, and action semantics across sources.

Real seeds and simulated visual augmentation. Real-robot trajectories supply the original task instructions, motions, and available observations, and anchor the scene and sensor calibration. Starting from the aligned assets, the simulation branch varies object appearance or instances and background visual elements. Tacinf-Tabero replays the task under these variations and controlled force regimes, producing synchronized visual observations, tactile images, marker fields, contact forces, and action targets. Background appearance changes leave the collision model unchanged; replacing an interactive object's geometry requires a corresponding collision asset and a new contact rollout. Tactile signals and force labels therefore come from replay in the varied scene, rather than being copied onto edited images. Task validity and contact consistency are checked before a replay is retained.

Spatial augmentation with UMI. Tacumi extends spatial coverage with human demonstrations. Objects are placed at different positions and orientations and approached along different paths. The shared gripper records the resulting changes in contact and motion together, adding physical spatial variation to the simulated visual and force-level variations. This branch collects new demonstrations of the task variants; changing an object's pose is not treated as an image-only transformation of an existing action sequence. Tracked hand poses and sensor observations are expressed through the common task and observation–action interfaces before mixing.

Joint VTLA post-training. The real, simulated, and UMI trajectories are synchronized, normalized, and sampled into mixed batches to jointly post-train a single pretrained Tacinf-VTLA policy with the action and fingertip-force supervision in Section 2.2. The source sampling ratio and task/force-level coverage are explicit training settings. Simulation contributes scalable visual and contact variation, real trajectories anchor physical observations, and UMI demonstrations extend object-placement and approach coverage. Held-out evaluation scenes and trials remain outside this training pool.

Closing the system loop. The post-trained policy is deployed with the same tactile gripper and hybrid controller. Physical execution reveals task configurations and contact conditions for further real-robot or UMI collection, which can seed another round of asset variation, tactile replay, and mixed post-training. This is a data-collection and post-training loop across iterations; the controller's within-trial tactile feedback remains the separate fast

execution loop. The co-training study in Section 3.8 specifies how to test the benefit of the combined data; numerical gains from mixed post-training or repeated collection rounds are not claimed in the current results.

3. Evaluation

The experiments in Sections 3.1–3.3 are based on the Tabero benchmark data, whereas the experiments from Section 3.4 onward are based on the Taco data pipeline; the controller ablation in Section 3.7 uses a mixture of both. In total, Tabero provides 18 task conditions and Taco provides 9 task conditions; the tasks cover the two categories introduced in the method section, namely following force-related language instructions and manipulating special objects. Over these tasks we ran over 10k simulated evaluations and over 2k real-robot evaluations on roughly a dozen tactile foundation models. The hardware consists of one XArm arm equipped with an Xense-based tactile gripper and its UMI counterpart for data collection, together with 4 H20 and 2 A6000 GPUs as the compute devices.

3.1. Cross-Platform Data Validation

To validate the effectiveness of the Tacinf-Tabero data migration method, we select LIBERO, a representative MuJoCo-based open-source dataset, as the data source. We validate the fidelity of data migration from MuJoCo to Isaac Lab by evaluating both task success rates and distributional consistency. Specifically, we select four subtasks from the LIBERO benchmark suite and compare the success rates of the original MuJoCo-based dataset with those of our replayed version in Isaac Lab. Fig. 2 illustrates the range of task scenarios our simulation platform can provide. In subsequent experiments, we only used a subset of these configurations and did not randomize the visual modality.

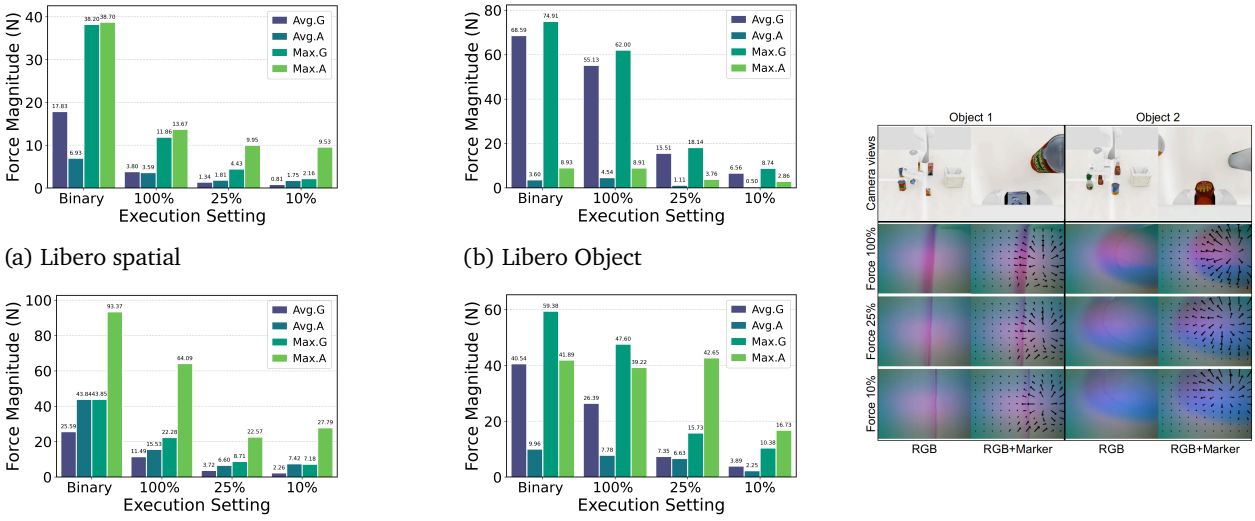
When using the same robot kinematics and control policy as in the original dataset, our baseline configuration yields a success rate distribution that closely matches that reported in OpenVLA Kim et al. (2025b). However, replacing the standard parallel-jaw end effector with a tactile-sensor-integrated gripper on the same Franka arm introduces mechanical differences that lead to a measurable drop in success rates. This degradation is especially pronounced in tasks requiring delicate manipulation, where lower grip forces strongly correlate with reduced success. The results shown in Table 1 highlight the sensitivity of contact-rich tasks to end-effector design and force regulation.

Table 1: Cross-platform data validation: task success rates across four LIBERO subtasks. We compare the original MuJoCo dataset, our replay in Isaac Lab with identical robot configuration (denoted Isaac), and our modified setup with a tactile-equipped Franka gripper at 100%, 25%, and 10% force scaling (denoted T-100, T-25, and T-10).

Task	MuJoCo	Isaac	T-100	T-25	T-10
Spatial	0.86	0.83	0.42	0.24	0.07
Object	0.91	0.77	0.84	0.87	0.73
Goal	0.76	0.78	0.55	0.44	0.30
LIBERO-10	0.86	0.66	0.60	0.47	0.32
Average	0.85	0.76	0.60	0.50	0.36

3.2. Tactile-Aware Task Design

We verify whether our data collection framework effectively expands the distribution of interaction forces. We compare a baseline using binary gripper control against our approach, which explicitly sets different force parameters during execution; results are shown in Fig. 7. Specifically, we define 100% grip force as the “firm”



(a) Libero spatial

(b) Libero Object

(c) Libero Goal

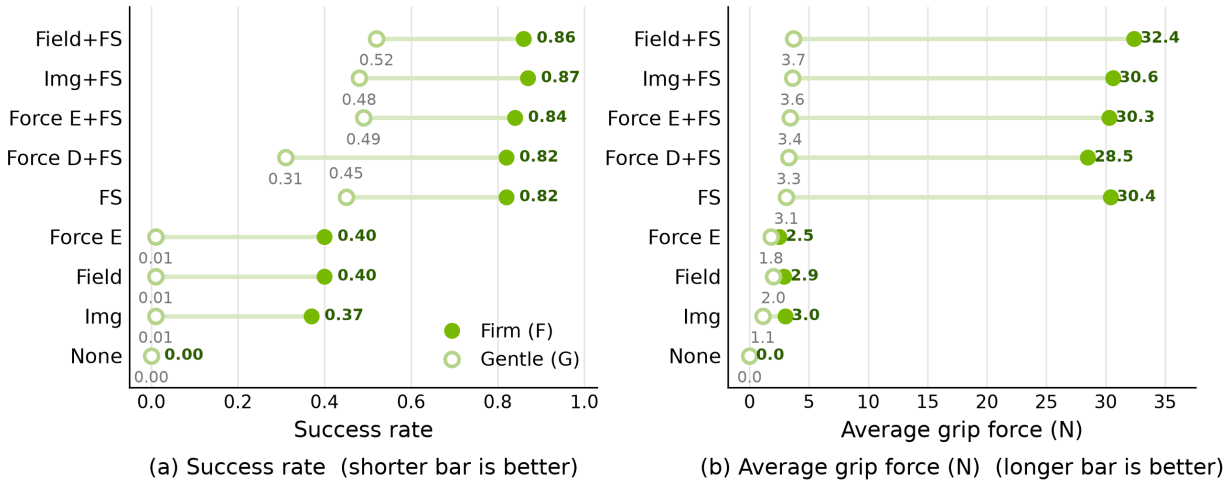
(d) Libero 10

Figure 7: **Force distribution and tactile observations at different force magnitudes. Left:** force distributions under various control modes across different task suites. "Binary" represents the binarized control commands applied by a non-tactile gripper; "100%", "25%", and "10%" indicate the force distributions of a tactile gripper under different force settings, with "100%" matching the non-tactile gripper. **Right:** tactile images and corresponding camera images at different force magnitudes. The left two columns represent the first category of objects and the right two columns the second; RGB denotes the simulated tactile RGB image, and RGB+Marker overlays the simulated marker motion field on the RGB image. In the camera views, the left view is from the third-person camera and the right view is from the wrist-mounted camera.

condition, modifying the original instruction with the adverbs "firmly" and "tightly", and 25% and 10% as the "gentle" conditions, annotated with "gently" and "softly". In our experiments, we set $K_p \in \{2000, 500, 200\}$ N/m and $K_d \in \{100, 25, 10\}$ N·s/m to correspond to 100%, 25%, and 10% force levels, respectively, where the 100% setting follows the default parameters for the Franka robot in Isaac Sim. For the same task, we present tactile images, force fields, and contact force measurements at comparable contact stages under these distinct force settings in Fig. 7, demonstrating clear multimodal variations aligned with the intended interaction intensity.

Furthermore, we observe that under the extreme 10% force setting, the retention rate of most tasks is relatively low. Appendix C gives the formal definition of the retention rate. Therefore, in subsequent ablation tests, we constructed a Tabero subset to analyze the policy's performance under extreme conditions. This subset includes 9 tasks from the Object dataset, each executed under two force conditions specified by linguistic adverbs. The composition of the dataset is presented in Appendix B.1. We collect three datasets over the task group, differing only in force magnitude and gripper actuation strategy. **Dataset A** uses continuous control with "gentle" forces at 25% of the corresponding "firm" level. **Dataset B** also employs continuous control but reduces the force to 10%, representing an extreme low-force regime where slippage is likely. **Dataset C** uses the same 10% force level but switches to binary open/close commands, with only discrete gripper states logged.

During evaluation, the arm and gripper operate with impedance parameters corresponding to 100% force scaling. The Tacinf-VTLA policy predicts target contact forces conditioned on language and visual inputs; these targets are tracked by a closed-loop admittance controller that modulates impedance during execution. We adapt a base VLA model using LoRA to incorporate tactile marker fields (Dataset A and B), while a vision–language-only variant is trained on Dataset C for ablation. As shown in Table 2, removing tactile feedback leads to complete failure in force modulation, highlighting its critical role in gentle manipulation. Furthermore, the



(a) Success rate (shorter bar is better) (b) Average grip force (N) (longer bar is better)

Figure 8: **Ablation study on tactile modalities.** (a) Success rates under firm (F) and gentle (G) force, where a shorter bar (a smaller firm–gentle gap) is better; (b) average grip force (N) under firm and gentle force, where a longer bar (a larger firm–gentle separation) is better. Solid dots mark firm, hollow dots gentle. None = no tactile input; Img = tactile image input; Field = force field input; Force E = force input via MLP encoder; Force D = force input via decoder; FS = force supervision loss enabled.

sharp drop in success rate from 25% to 10% force scaling reflects the increased difficulty of maintaining stable contact under ultra-gentle constraints, underscoring that Dataset B represents a substantially more challenging regime for gentleness-aware policies. Appendix C.2 reports the corresponding language-to-force trend on this dataset, including the out-of-domain adverbs.

Table 2: Training results on the three datasets. Each cell gives the firm / gentle value, where the gentle condition is 25% for Dataset A, 10% for Dataset B, and 10% with binary gripper commands for Dataset C.

Metric	A 100% / 25%	B 100% / 10%	C 100% / 10%
SR	0.87 / 0.79	0.86 / 0.52	0.86 / 0.87
AG	31.3 / **8.5**	32.4 / **3.7**	35.8 / 34.6
MG	61.3 / 19.9	62.7 / 11.4	65.7 / 65.0

3.3. Ablation and Comparison of VTLA

To compare and conduct ablation experiments on different tactile injection methods and force control strategies, we evaluate tasks from **Dataset B**. Among them, Force E+FS is designed based on Huang et al. (2025c), Img is designed based on Zhang et al. (2026), and Force D+FS is designed based on Zhang et al. (2025b). These are adaptations of the cited design principles within our shared backbone and training protocol, rather than reproductions of the original systems or their reported results. Except where the ablation architecture requires a different tactile interface, all models were fine-tuned via LoRA Hu et al. (2022) with an identical set of hyperparameters; detailed parameters are reported in Appendix A.

Fig. 8 shows that without tactile input the baseline control policy fails completely. This confirms that gentle grasping requires not only compliant actuation but also sensory awareness of interaction forces. When tactile inputs such as images or force fields are provided, the policy gains basic force modulation ability and achieves nontrivial success. Adding explicit force supervision enables precise force prediction and substantially improves performance under gentle conditions. The best results come from combining rich tactile representations like marker fields with force supervision, highlighting their complementary roles in gentleness-aware manipulation.

3.4. Taco Data Alignment

We use the data-replay pipeline introduced by Tabero to obtain, from single-distribution real-robot data, simulated Taco data with "firm" and "gentle" force distributions, and evaluate the consistency between the two domains. To complement task-level validation, we measure the consistency along three channels: visual images, tactile images, and action labels. Visual consistency is quantified by the CLIP similarity of the third-person and wrist camera views; tactile consistency by the structural similarity (SSIM) and the mean absolute error (MAE) between simulated and real tactile images; and action consistency by the MAE of the action labels. Table 3 reports these metrics for three firm-force tasks. The results show that the wrist and third-person camera views are well aligned (both CLIP similarities above 0.75); the tactile signals further agree in SSIM and MAE, and the actions are also well aligned, demonstrating the data reliability of the Taco replay pipeline.

Table 3: Sim–real data consistency across visual, tactile, and action channels.

	Third-person CLIP	Wrist CLIP	Tactile SSIM	Tactile MAE	Action MAE
Firm Task 1	0.781	0.770	0.941	0.0160	0.0708
Firm Task 2	0.785	0.754	0.931	0.0182	0.0685
Firm Task 3	0.794	0.766	0.932	0.0196	0.0733

3.5. Policy-level Zero-shot Sim2Real Force-level Transfer

We test whether Taco supports policy-level zero-shot transfer of language-conditioned force levels from simulation to the physical robot. Each policy is initialized from a pretrained VLA and receives task-specific adaptation using simulated data only; after Real2Sim calibration, the same adapted policy, language conditions, and controller are executed in the aligned simulation (Sim) and directly on the physical robot (Sim2Real), without real-robot task demonstrations, policy fine-tuning, or test-time adaptation. The force regimes are changed during simulation data generation, while the robot evaluation measures whether the corresponding force ordering is preserved. Real tactile observations used for simulator and observation-mapping calibration are separate from policy optimization. Evaluation uses frozen calibrated assets and controller settings, with evaluation scenes and random seeds separated from training. Following the paired evaluation protocol of PolaRiS Jain et al. (2026), the two evaluations share the same task set, language instructions, controller settings, and termination criteria. For every policy we report two complementary metrics in both domains. The first is the success rate averaged over the firm (F) and gentle (G) conditions, which summarizes overall task competence. The second is the force-magnitude separation $\Delta = \mathrm{F\,AG} - \mathrm{G\,AG}$, the difference between the firm and gentle average grip force: a larger Δ indicates that the policy scales its grip force more strongly with the commanded force level, i.e., that it follows the force magnitude more faithfully. Policies that fail under both conditions report a success rate of 0 in both domains, and their Δ is reported as 0.

Table 4 reports the results. Without tactile input or tactile supervision, the vision-only variants did not solve the evaluated tasks under our protocol. In simulation, all tactile policies achieve substantial force separation; on the physical robot, the absolute separation shrinks, reflecting the residual sim-to-real gap in force response, but the policies with the strongest force-magnitude following are consistent across domains: Force E+FS attains the largest separation in simulation and the second-largest on the robot (2.3047 N), while Field+FS (pi0.5, state) ranks first on the robot (2.3955 N). Notably, Field+FS (pi0) reaches a high success rate on the robot (95.0%) yet barely differentiates the two force levels ($\Delta = 0.6146$ N), whereas Field+FS (pi0.5, no state) achieves the best trade-off with a mean success rate of 96.665% and a clear force separation ($\Delta = 1.4585$ N). Absolute success rates still differ between the two domains, yet both domains agree on the central conclusion: tactile input with force supervision is necessary for force-aware manipulation, and combining tactile representations with force supervision yields the strongest force-magnitude following. Thus the result concerns transfer of

force-level ordering and modulation, rather than equality of absolute force measurements across hardware.

To bound the remaining transfer gap, Table 5 reports the raw firm/gentle success rates and grip-force statistics of the same two policies trained on real-robot data and evaluated on the physical robot (Real2Real), alongside their Sim2Real counterparts. Real-data training raises the success rate of Img+FS to 100% and roughly doubles its measured grip-force scale, yet its firm average grip force falls slightly below the gentle one (13.0747 N vs. 14.0468 N), indicating that this configuration no longer differentiates the two commanded force levels; Field+FS keeps a positive firm–gentle separation in both domains (F AG 9.691 N vs. G AG 9.1864 N) at a nearly identical success rate.

Table 4: Taco Sim2Real validation of simulated policies in the real and simulated domains. Mean SR is the success rate averaged over the firm (F) and gentle (G) conditions; Δ is the force-magnitude separation $\mathrm{F\,AG} - \mathrm{G\,AG}$. For both metrics, larger values are better. A dash denotes a cell not evaluated in that domain. Simulation results for the remaining Img+FS ($\pi_{0.5}$) variants are reported in Appendix C.3.

	Sim2Real		Sim	
Model	Mean SR (%) ↑	Δ (N) ↑	Mean SR (%) ↑	Δ (N) ↑
None ($\pi_{0.5}$)	0	0	0	0
Img (π_0, w=0)	0	0	0	0
Force D+FS (π_0)	81.14	1.6407	13.65	11.52
Force E+FS (π_0)	94.165	2.3047	34.7	**18.57**
Field+FS (π_0)	95.0	0.6146	57.0	16.33
Field+FS ($\pi_{0.5}$, state)	82.195	**2.3955**	71.35	16.49
Field+FS ($\pi_{0.5}$, no state)	**96.665**	1.4585	–	–
Img+FS ($\pi_{0.5}$, no state, $H = 1$)	84.44	1.1142	**95.65**	15.39

Table Notes: Sim2Real evaluates policies after simulation-only task adaptation directly on the physical robot; Sim evaluates the same adapted policies in the aligned simulation. Mean SR = (F SR + G SR)/2, the mean of the firm and gentle success rates; $\Delta = \mathrm{F\,AG} - \mathrm{G\,AG}$, the difference between the firm and gentle average grip force (N), indicating how strongly the policy follows the commanded force magnitude. state/no state = with/without proprioceptive state input; w=0 = tactile-loss weight set to zero; $H = 1$ = the composite tactile image contains only the current contact state (no historical frames). None and Img achieve a success rate of 0 in both conditions and domains, with Δ set to 0. The other Img+FS ($\pi_{0.5}$) variants (state, no state, state $H = 1$) are evaluated in simulation only; see Appendix C.3.

Table 5: Real2Real reference of the two best policies: raw firm/gentle success rates and grip-force statistics of the same models trained on real-robot data and evaluated on the physical robot (Real2Real), alongside their Sim2Real counterparts in Table 4. F SR/G SR = success rates under firm/gentle force; F AG/G AG = average grip force (N); F MG/G MG = maximum transient grip force (N).

Model	F SR	G SR	F AG (N)	G AG (N)	F MG (N)	G MG (N)
Sim2Real						
Img+FS ($\pi_{0.5}$, no state, $H = 1$)	0.9198	0.769	6.9236	5.8094	7.5757	6.3726
Field+FS ($\pi_{0.5}$, no state)	0.95	0.9833	6.4179	4.9594	7.0031	5.4184
Real2Real						
Img+FS ($\pi_{0.5}$, no state, $H = 1$)	1	1	13.0747	14.0468	14.9271	15.3706
Field+FS ($\pi_{0.5}$, no state)	0.9667	0.9667	9.691	9.1864	10.9326	10.2224

3.6. Taco Policy Performance Consistency

Following SIMPLER and PolaRiS Jain et al. (2026); Li et al. (2025b), we quantify the faithfulness of the simulated evaluation with the Pearson correlation and the Mean Maximum Rank Violation (MMRV) between the Sim2Real and Sim scores, computed across the seven policy variants evaluated in both domains: the five tactile variants with paired Sim and Sim2Real scores, together with the two zero-scoring vision-only baselines. The suite spans two pretrained VLA backbones and several tactile input/force-supervision configurations, so this is a cross-model consistency test rather than a single-checkpoint demonstration. Because we assess both task competence and force-magnitude following, each model's combined score averages its Mean SR and its Δ after min–max normalization within each domain, so that both metrics jointly determine the ranking. Fig. 9 reports $r = 0.847$ and MMRV$= 0.19$ for the combined score; using success rate alone gives $r = 0.702$ and MMRV$= 0.069$. The combined score therefore tests the harder claim that simulation preserves both task and force-level ordering. The largest rank inversions involve Force E+FS and Img+FS (H=1), and likely reflect grasp posture and hardware reliability in addition to the sensor representation.

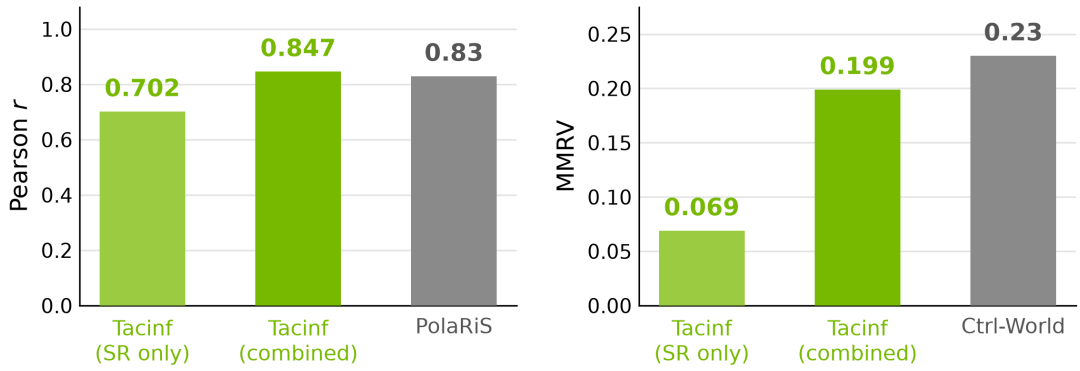
It is worth stressing the regime of this validation: after pretrained-VLA initialization, task-specific policy adaptation uses simulated data only; no real-robot task demonstrations, policy optimization, or test-time adaptation are used. We deliberately deviate from the PolaRiS comparison protocol, in which policies pretrained on large real-robot datasets are co-finetuned on batches that mix new simulation data with the original real pretraining data, because our goal is to isolate the reliability of the simulated data and its force-level labels. Under this stricter sim-data-only adaptation setting, our combined-score agreement $r = 0.847$ is numerically close to the average Pearson $r = 0.83$ reported by PolaRiS in its published RSS version with sim + real co-training. The protocols, model pools, and sample sizes differ (seven policy variants in our analysis versus four DROID VLAs in PolaRiS), so we do not interpret the numerical difference as a direct improvement. The more specific evidence is that one calibrated simulation preserves relative force-level behavior across multiple tactile-policy variants without real-robot policy optimization. For context, PolaRiS reports an MMRV of 0.23 for Ctrl-World, an action-conditioned video world model trained on real-robot videos; this value is not a direct result from our protocol. Taco therefore supports a narrowly defined zero-shot force-level transfer claim, while the fine-grained ranking among capable policies motivates the Real2Sim alignment and co-training refinements described below.

3.7. Effectiveness of Hybrid Controller

To isolate the contribution of the low-level controller from the high-level policy, we evaluate on a mixture of the Tabero **Dataset A** and Taco data, where task success is largely insensitive to force variations. The hyperparameters of our controller are presented in Appendix B. We conduct four ablation studies on the gripper controller: (a) full force with hybrid control, (b) reduced force with hybrid control, (c) reduced force without feedforward term, and (d) reduced force without admittance component. Results in Fig. 10 demonstrate that both feedforward and admittance terms are essential for accurate force tracking.

For the arm controller, we focus only on gripper force feedback, as the applied interaction forces in our tasks are small and have limited influence on arm motion. Consequently, we highlight the effectiveness of tactile-based grip force regulation, which is critical for successful grasping.

Planned real-robot task expansion. We will transfer the best Tacinf-VTLA checkpoint identified in the preceding evaluations to new real-robot tasks while keeping the same Tacumi sensing path and closed-loop controller. The protocol will specify whether each task is evaluated zero-shot or with minimal adaptation before data collection; the setting and adaptation budget will be reported with the completed results. The task set is chosen to exercise contact quantities that complement the grasping experiments: blackboard wiping requires controlled downward normal pressure, card swiping requires a stable normal applied force, and bottle-cap twisting requires torsional torque. For every task we will report success, target and measured contact quantities,



(a) Pearson correlation (higher is better) (b) Mean Maximum Rank Violation (lower is better)

Figure 9: **Cross-domain agreement of simulated evaluations.** (a) Pearson correlation between simulated and real-robot scores, higher is better; (b) Mean Maximum Rank Violation (MMRV), lower is better. Green bars are our policies after simulation-only task adaptation: the success-only score gives $r = 0.702$, MMRV$= 0.069$, while the combined task-and-force score gives $r = 0.847$, MMRV$= 0.19$. The agreement statistics are computed across the seven policy variants evaluated in both domains following the PolaRiS Jain et al. (2026) protocol (PolaRiS itself uses four DROID VLAs); the combined score averages the per-domain min–max normalized Mean SR and Δ, so both metrics jointly determine the ranking. For context, the published RSS 2026 version of PolaRiS reports an average Pearson $r = 0.83$ and an MMRV of 0.23 for its evaluation of Ctrl-World Guo et al. (2026); the latter value is reported by PolaRiS, not by the original Ctrl-World paper. Ctrl-World is an action-conditioned video world model whose evaluation rollouts are judged by humans. The literature values use different task suites and scoring protocols and are contextual references, not directly comparable benchmark results.

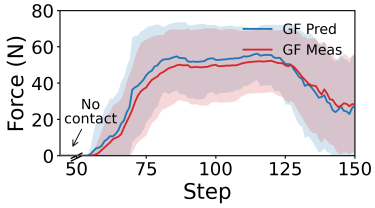
tracking error, peak load, and slip or failure rate (Table 6). The experiment tests whether the policy and controller generalize their force-aware behavior to new contact dynamics, rather than only to the tasks used for training.

Table 6: Real-robot task expansion with force- and torque-sensitive contact objectives. Target, measured, and peak quantities use N for forces and N·m for torsional torque, as specified in the second column.

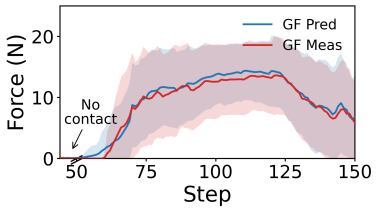
Task	Dominant Contact Quantity	# Trials	SR	Mean Target	Mean Measured	Tracking MAE	Peak Quantity	Slip/Failure Rate
Blackboard wiping	Downward normal pressure (N)	–	–	–	–	–	–	–
Card swiping	Normal applied force (N)	–	–	–	–	–	–	–
Bottle-cap twisting	Torsional torque (N·m)	–	–	–	–	–	–	–

3.8. Real–Simulation–UMI Co-Training

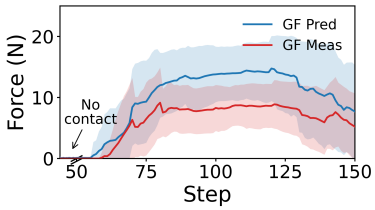
This is a planned experiment for the augmentation and joint post-training loop in Section 2.4.3; no numerical co-training result is claimed in the current version. We will evaluate whether adding Taco trajectories to real-robot and UMI data improves a single shared tactile policy. The policy will be trained directly on mixtures of $\mathcal{D}_{\text{real}}$, $\mathcal{D}_{\text{sim}}$, and $\mathcal{D}_{\text{umi}}$ using the same observation, action, timestep, and normalization interfaces; we will not train separate domain-specific policies followed by distillation. Table 7 organizes the evaluation over the six original tasks with object-material and position randomization and background disturbance, comparing five training mixes: real-only, real+umi (1:2), sim+umi (10:2), real+sim (1:10), and real+sim+umi (1:2:10). For each mix we will report the firm/gentle success rates and the average and maximum grip forces on held-out real scenes. This design extends the simulation-data co-training perspective in PolaRiS Jain et al. (2026) to a three-source tactile interface.



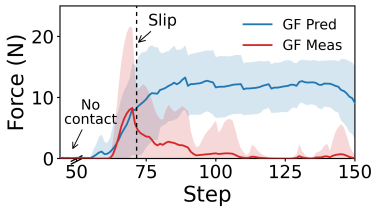
(a) 100% force



(b) 25% force



(c) 25%, no feed-forward



(d) 25%, no admittance

Figure 10: **Ablation study on gripper force control.** GF stands for gripper force. In Tabero Object task 1, the predicted force is shown in blue and the measured force in red: (a) 100% force, (b) 25% force, (c) 25% force without feedforward term, and (d) 25% force without admittance control. Slip stands for object dropping.

Our working hypothesis is that the three sources provide complementary coverage rather than interchangeable samples: simulation supplies object/background visual variation, controlled force-level variation, and tactile replay at scale; real-robot data anchors sensor, actuator, and scene statistics; and UMI data contributes human-guided contact trajectories under varied object positions, orientations, and approach motions. We therefore expect the mixed policy to have the best chance of improving held-out real force-level separation when the sampling ratio is controlled, while a simulation-heavy mixture may increase force coverage at the cost of residual observation mismatch and a real-only mixture may under-cover rare force levels. These are testable hypotheses, not assumed outcomes. The analysis will compare force-level separation, success under firm/gentle instructions, peak-force suppression, and the Sim–Real consistency metrics under the same held-out scenes; all numerical cells remain to be filled after training.

Table 7: Planned real–simulation–UMI co-training results under different training mixtures on the six original tasks with object-material and position randomization and background disturbance. F SR/G SR = success rates under firm/gentle force; F AG/G AG = average grip force (N); F MG/G MG = maximum transient grip force (N). Numerical entries are intentionally left blank until training is complete.

Training Mix	F SR	G SR	F AG (N)	G AG (N)	F MG (N)	G MG (N)
only real	–	–	–	–	–	–
real+umi (1:2)	–	–	–	–	–	–
sim+umi (10:2)	–	–	–	–	–	–
real+sim (1:10)	–	–	–	–	–	–
real+sim+umi (1:2:10)	–	–	–	–	–	–

4. Related Work

4.1. Simulation Platforms and Synthetic Data

Robot simulation provides reproducible environments for policy learning and evaluation. Early infrastructures such as robosuite, LIBERO, and CALVIN established reusable manipulation tasks and language-conditioned evaluation protocols Liu et al. (2023); Mees et al. (2022); Zhu et al. (2020). Recent platforms extend these

capabilities through household scene diversity in RoboCasa, GPU-parallel simulation in ManiSkill3 and Isaac Lab, and unified assets and interfaces in RoboVerse Geng et al. (2025); Nasiriany et al. (2024); NVIDIA et al. (2025b); Tao et al. (2025). MuBlE couples MuJoCo dynamics with Blender rendering for tasks requiring both visual and physical reasoning Nazarczuk et al. (2025). At the data-generation level, RoboTwin and RoboTwin 2.0 combine task construction and demonstration generation for bimanual manipulation, while SkillMimicGen reuses demonstrations through skill-level generation Chen et al. (2025); Garrett et al. (2025); Mu et al. (2025). GraspVLA further demonstrates the use of large-scale synthetic action data for policy pretraining Deng et al. (2025). These approaches motivate our reuse of existing trajectories, but trajectory replay alone does not establish that contact forces or tactile observations transfer across simulators.

A complementary direction reconstructs real environments for simulation-based policy evaluation. SIMPLER studies visual and control gaps and evaluates whether simulated performance predicts real-robot behavior; PolaRiS combines neural scene reconstruction with simulation-data co-training Jain et al. (2026); Li et al. (2025b). SimFoundry automates scene reconstruction and generates task-relevant scene variations, while Seed3D 2.0 provides simulation-oriented 3D asset generation Gu et al. (2026); Ranawaka et al. (2026). Ctrl-World instead uses an action-conditioned generative world model for policy evaluation and improvement Guo et al. (2026). These works establish precedents for real-to-sim reconstruction, data mixing, and cross-domain policy ranking. Tacinf-Taco adapts this direction to tactile observations and force-conditioned evaluation.

Tactile simulation datasets and benchmarks are also emerging. VTDexManip supports visual–tactile pre-training and dexterous manipulation, SafeDiff evaluates force-aware interaction using SafeDoorManip50k, and UniVTAC unifies multi-sensor tactile synthesis, representation learning, and manipulation benchmarking Chen et al. (2026); Liu et al. (2025b); Wei et al. (2025). SoftVTBench pairs visuotactile observations with independent deformation ground truth for evaluating deformable-object interaction quality Jing et al. (2026). Thus, neither tactile data synthesis nor a unified tactile platform is itself new. Tacinf-Tabero focuses on converting existing manipulation trajectories into synchronized vision–tactile–language supervision and on evaluating instruction-dependent interaction forces alongside task completion.

4.2. Tactile Simulation Methods

Vision-based tactile simulation must account for contact geometry, sensor deformation, and image formation. TACTO couples physics-engine contact information with efficient optical rendering Wang et al. (2022). Taxim calibrates an example-based optical mapping and models marker motion using elastic deformation and superposition; FOTS provides fast optical and marker-motion approximations for tactile-motor learning Si and Yuan (2022); Zhao et al. (2024). These methods offer efficient sensor-observation models, although their calibration assumptions and approximations must be distinguished from a complete elastomer dynamics model. TacEx integrates soft-body contact simulation with Taxim and FOTS in Isaac Sim, and TacSL supports parallel visuotactile simulation and learning Akinola et al. (2025); Nguyen et al. (2024).

Physics-based approaches pursue more explicit contact mechanics. DiffTactile combines differentiable sensor and object simulation for system identification and trajectory optimization, while Taccel integrates contact and rigid-body dynamics in a GPU-parallel tactile simulation platform Li et al. (2025c); Si et al. (2024). HydroShear additionally models path-dependent shear accumulation and stick–slip transitions Dang et al. (2026). Learned observation transfer offers a complementary route: SimShear generates shear-aware tactile images from simulated observations and shear conditioning, and Cross-Sensor Touch Generation translates tactile observations between sensor designs Freud et al. (2025); Rodriguez et al. (2025). Applications such as AnyRotate, UniTac2Pose, and VT-Refine demonstrate how simulated touch can support in-hand manipulation, pose estimation, and policy refinement Huang et al. (2025b); Wu et al. (2025b); Yang et al. (2025). Together, these studies expose a tradeoff between contact fidelity, calibration effort, and simulation throughput, and they establish tactile sim-to-real transfer as an important evaluation target Freud et al. (2025); Yang et al.

(2025). Following the modular integration exemplified by TacEx, Tacinf uses Taxim for optical observations and FOTS for marker motion; our contribution concerns their calibrated integration into a scalable data and evaluation pipeline and the narrower evaluation of language-conditioned force-level transfer across multiple policy variants, rather than a new sensor-mechanics solver or a generic first claim about tactile sim-to-real deployment.

4.3. Tactile Data Collection

Large robot datasets such as DROID demonstrate the value of diverse real-world demonstrations, while UMI moves demonstration capture to handheld devices Chi et al. (2024); Khazatsky et al. (2024). FastUMI further develops a scalable, hardware-independent collection interface Zhaxizhuoma et al. (2025). For tactile learning, sensor hardware and observation alignment are equally important: GelSight provides high-resolution contact geometry and deformation measurements, PolyTouch combines complementary sensing modalities, and 3D-ViTac fuses distributed tactile measurements with visual geometry Huang et al. (2025a); Yuan et al. (2017); Zhao et al. (2025a). Their differing sensor layouts and signal types create challenges for sharing demonstrations across embodiments.

Recent robot-free systems directly address tactile data acquisition. ViTaMIn uses a compliant handheld gripper with visuotactile sensing and operator-perceptible force feedback; ViTaMIn-B extends collection to bimanual manipulation with compliant sensors and tracked controllers Li et al. (2025a); Liu et al. (2025a). FreeTacMan uses a wearable gripper for synchronized visual–tactile demonstrations, whereas exUMI combines modular tactile sensing, calibrated state acquisition, and action-aware tactile pretraining Wu et al. (2025a); Xu et al. (2025b). TacUMI integrates visuotactile sensing, force–torque measurements, and pose tracking for multimodal collection and task segmentation. UMI-FT records finger-level wrenches and learns position, grasp-force, and stiffness commands for compliant execution Cheng et al. (2026a); Choi et al. (2026). These systems provide several forms of physical feedback; comparing them requires distinguishing recorded tactile observations, direct mechanical feedback to the demonstrator, and actively rendered haptic feedback.

Human-to-robot transfer also has multiple established solutions. MimicTouch learns from human tactile demonstrations with residual adaptation, DexUMI reduces embodiment differences through an exoskeleton and visual inpainting, and KineDex uses kinesthetic teaching to collect tactile-informed demonstrations Xu et al. (2025a); Yu et al. (2025a); Zhang et al. (2025a). TactAlign aligns tactile representations across human and robot embodiments Wi et al. (2026). Tacinf-Tacumi explores a complementary implementation: a motor-torque feedback channel and the same tactile gripper hardware for handheld collection and robot deployment. This design targets consistency of contact observations and actuation, while recognizing that shared gripper morphology alone cannot eliminate all collection-to-deployment differences.

4.4. Tactile Foundation Models

Tactile foundation models encompass transferable perception backbones as well as policies conditioned on touch. Learning to Jointly Understand Visual and Tactile Signals and UniTouch align tactile observations with visual or multimodal representations Li et al. (2024); Yang et al. (2024). Sparsh studies self-supervised tactile pretraining and introduces TacBench; Transferable Tactile Transformers uses a shared backbone with sensor-specific and task-specific modules Higuera et al. (2025a); Zhao et al. (2025b). Sensor-Invariant Tactile Representation and AnyTouch address cross-sensor transfer, with AnyTouch explicitly combining static and dynamic tactile information Feng et al. (2025); Gupta et al. (2025). TacX extends pretraining beyond tactile images to multiple sensing modalities, while AnyTouch 2 emphasizes temporal deformation and force-aware dynamics Feng et al. (2026); Higuera et al. (2025b). These advances establish that scalable touch learning depends on sensor diversity, temporal information, and appropriate transfer evaluations, not simply on treating a tactile image as another camera view.

Tactile–language models connect contact observations to semantic reasoning. Octopi and Octopi-1.5 use tactile inputs for object-property inference and language-grounded reasoning Yu et al. (2024, 2025b). On the action side, OpenVLA, OpenVLA-OFT, π_0, $\pi_{0.5}$, RDT-1B, and GR00T N1 provide generalist policy architectures and training recipes that motivate tactile extensions Black et al. (2025a,b); Kim et al. (2025a,b); Liu et al. (2025c); NVIDIA et al. (2025a). TLA and VTLA study language-conditioned contact-rich manipulation, including insertion and preference-based learning Hao et al. (2026); Zhang et al. (2026). TA-VLA examines torque-aware policy design; OmniVTLA aligns heterogeneous tactile signals with semantic representations; VLA-Touch introduces tactile feedback at planning and execution levels Bi et al. (2026); Cheng et al. (2026b); Zhang et al. (2025b). Tactile-VLA already combines tactile-conditioned prediction with hybrid position–force control Huang et al. (2025c). Related visuotactile policies such as RoboPack and Reactive Diffusion Policy use touch for dynamics inference or fast reactive control Ai et al. (2024); Xue et al. (2025).

Tacinf-VTLA builds on these representation and control directions. It studies alternative tactile inputs, predicts fingertip-force targets, and couples a pretrained policy to separate grip-force and applied-wrench feedback loops. Accordingly, we position it as an instantiation and evaluation of force-aware tactile policy learning within Tacinf, rather than claiming that tactile tokenization, multimodal fusion, or hybrid force–position control are introduced here for the first time.

5. Conclusions

We present Tacinf, an integrated tactile infrastructure for language-conditioned force-level following. Tacinf-Tabero turns open trajectories into calibrated multimodal data with controlled force regimes and interaction-quality metrics. Tacinf-VTLA instantiates tactile input adapters, force supervision, and a two-loop force–position controller within an existing VLA architecture, so language conditions can select and regulate different force levels rather than only a binary gentle/firm behavior. Tacinf-Tacumi provides a handheld UMI device with the same tactile gripper used for deployment. Tacinf-Taco performs task-aligned Real2Sim construction and evaluates policies after simulation-only task adaptation on the physical robot without real-robot task demonstrations, policy fine-tuning, or test-time adaptation; across multiple policy variants, it preserves both task and force-level ordering under the paired Sim/Sim2Real protocol. Together, these components define a reusable path from open trajectories and force-conditioned synthetic data to tactile policy evaluation on hardware.

The demonstrated contribution is the shared force semantics and zero-shot evaluation path. The broader three-source co-training interface is specified but its empirical benefit remains to be measured. **Limitations.** Our current framework does not jointly optimize for both task success and minimal interaction force. In ultra-gentle regimes, there exists an inherent trade-off between gentleness and reliability. The current quantitative evaluation also does not yet complete the Tacumi–Taco real–simulation–UMI co-training study or validate the full hardware path on new physical tasks. Future work can test the stated coverage and calibration hypotheses, explore reinforcement learning to balance success and force objectives, and evaluate robust Tacinf-VTLA deployment on real hardware.

A. Training and Interface Contract

The model variants share the backbone, optimizer family, action convention, and evaluation protocol. We report the reusable interface and claim-critical settings here; raw rollout logs, manufacturer specifications, random seeds, checkpoints, and full data manifests are released separately. The Xense values below describe the policy-visible observation interface rather than a mechanically verified sensor.

Table 8: Common training and action-interface settings.

Parameter	Value
Backbone family	pi0 (JAX)
Padded action dimension	32
Semantic action dimension	13
Fingertip force dimension	6
Tactile history length (H)	8 frames
Optimizer schedule	Cosine decay
Peak / final learning rate	2.5×10^{-5} / 2.5×10^{-6}
Training steps	50,000
Batch size	32
Force-loss weight	0.1 when enabled; 0 otherwise

Table 9: Tactile input variants used in the ablations. "Prefix" and "suffix" denote the injection location in the backbone; "FS" denotes force supervision.

Variant	Visual streams	Tactile stream	Adapter	FS
Img+FS	3	–	–	yes
Img	3	–	–	no
Force D+FS	2	suffix	MLP	yes
Force E	2	prefix	MLP	no
Force E+FS	2	prefix	MLP	yes
Field+FS	2	prefix	TCN	yes
Field	2	prefix	TCN	no
FS	2	–	–	yes
None	2	–	–	no

Adapter settings.

The force adapter is a two-layer MLP with swish activation. Its input is the $H \times 6$ history of fingertip forces; the expert width is 1024 for suffix injection and 2048 for prefix injection, with hidden width $2W$. The marker-field adapter is a two-layer causal TCN with kernel size 3, history $H = 8$, swish activation, and prefix width 2048. The marker input is represented as $(H + 1) \times N \times 2$, where the first frame is the undeformed reference layout. Table 9 lists the variants these adapters are used in.

Interface items to release.

The public data contract should specify, for each stream, units, frame convention, timestamp policy, normalization statistics, and missing-value handling. The current draft fixes the action and force dimensions above; the release manifest should fill the remaining fields before publication: camera frame rate *TBD*, policy/control frequency *TBD*, RGB-D resolution *TBD*, and the path to the normalization file *TBD*.

B. Controller and Task Protocol

The two-loop controller tracks the policy's pose and fingertip-force targets. The values in Table 10 are controller settings for the Hybrid+Tactile configuration and are not claimed to be hardware-independent defaults.

Table 10: Force–position hybrid-controller settings.

Parameter	Value
Grip feed-forward gain / deadzone	0.6 / 1.0
Cartesian position gains (x, y, z)	$(-10^{-4}, -10^{-4}, -10^{-4})$
Grip proportional gain / deadzone	0.0008 / 0.25
Force filter coefficient α	0.2
IK command / solver	pose / DLS
IK scale / position offset	1.0 / $[0, 0, 0.1034]$

B.1. Language-Conditioned Force Protocol

For each base instruction, force-related adverbs are inserted either before or after the verb phrase and the insertion position is randomized. The in-domain pairs are "firmly/tightly" for the high-force condition and "gently/softly" for the low-force condition. Unseen adverbs such as "lightly" and "forcefully" are reserved for the generalization test and never enter training. Table 11 gives example templates.

Table 11: Example instruction templates for force-language conditioning.

Template	Instruction
Prefix-high	*firmly pick up the alphabet soup and place it in the basket*
Suffix-high	*pick up the alphabet soup and place it in the basket tightly*
Prefix-low	*softly pick up the alphabet soup and place it in the basket*
Suffix-low	*pick up the alphabet soup and place it in the basket gently*

The Tabero subset contains the nine base tasks listed in Table 12. The force adverbs are applied to every task using the protocol above.

Table 12: Base tasks in the Tabero subset.

ID	Task description
0	pick up the alphabet soup and place it in the basket
1	pick up the cream cheese and place it in the basket
2	pick up the salad dressing and place it in the basket
3	pick up the bbq sauce and place it in the basket
4	pick up the tomato sauce and place it in the basket
5	pick up the butter and place it in the basket
6	pick up the milk and place it in the basket
7	pick up the chocolate pudding and place it in the basket
8	pick up the orange juice and place it in the basket

C. Supplementary Evaluation

C.1. Retention Summary

We define retention as the fraction of replay episodes that complete the task successfully, $R = N_s/N_t$, where N_s is the number of episodes that complete the task and N_t is the number of replayed episodes.

C.2. Semantic-Force Generalization

Table 13 reports the intended format for the in-domain and out-of-domain linguistic-adverb test. The draft values are retained for the current snapshot; they must be regenerated if the dataset, checkpoint, or controller changes.

Table 13: Semantic-force understanding on Dataset B. Force values are in N.

Adverb	SR	AG	AA	MG	MA
firmly, tightly	0.86	32.4	1.5	62.7	6.9
gently, softly	0.52	3.7	0.4	11.4	5.1
lightly (OOD)	0.61	14.5	0.8	40.8	7.6
forcefully (OOD)	0.64	18.1	0.9	49.7	6.3

The test is interpreted as a language-to-force trend, not as a calibration of absolute force. Report the number of episodes, random seeds, and confidence intervals in the release manifest; these fields are currently *TBD*.

C.3. Img+FS Variants

Table 14 reports the remaining Img+FS ($\pi_{0.5}$) variants under simulation.

Table 14: Additional Img+FS ($\pi_{0.5}$) variants under simulation.

Model	Sim SR (%)	Sim Δ (N)
Img+FS (state)	86.35	16.21
Img+FS (no state)	90.65	13.55
Img+FS (state, $H = 1$)	91.70	16.74
Img+FS (no state, $H = 1$)	95.65	15.39

D. Tactile Real2Sim Calibration

The calibration target is the policy-visible observation interface: contact onset, patch persistence, force scale, marker response, channel order, and temporal update semantics. It is not a claim of pixel-identical digital-twin fidelity.

D.1. Depth-to-Indentation Mapping

At each simulation step, the depth camera produces a height map $D(u, v)$, and we write $Z(u, v) = 1000\, D(u, v)$ in millimetres. Let $d_{\min} = \min_{u,v} Z(u, v)/1000$, d_{ref} be the camera-to-gel reference distance, and h_g the effective gel thickness. The indentation depth and the height map passed to Taxim and FOTS are

$$\delta = \max(0, d_{\min} - d_{\text{ref}})\,, \qquad h_{\text{in}} = \begin{cases} (h_g - \delta)\,1000, & \delta \le h_g, \\ 0, & \delta > h_g, \end{cases} \tag{13}$$

with h_{in} in millimetres and $Z'(u,v) = Z(u,v) - \min_{u,v} Z(u,v) - h_{\text{in}}$ before rendering.

D.2. Taxim and FOTS Interface

Taxim Si and Yuan (2022) maps local height-map gradients and a no-contact background to the tactile RGB observation. The calibration bundle contains the background, gradient-bin polynomial, gel reference map, and sensor/deformation settings; these files are released with the simulator configuration. The current branch uses RGB ordering, calibrated image-coordinate normalization, and no additional shadow term.

FOTS Zhao et al. (2024) receives the shifted height map and Taxim contact mask. Normal deformation determines contact centre and magnitude, while the task-frame transform supplies tangential motion and yaw. Marker displacement is propagated with

$$g_k(x,y) = \exp\left[-\lambda_k\left((x-c_x)^2 + (y-c_y)^2\right)\right]. \tag{14}$$

Exact camera geometry, pad dimensions, spatial-decay coefficients, and image-processing kernels are configuration-level details and are released with the simulator branch. They are intentionally not repeated here because they do not define a new scientific claim and may change with the calibrated sensor asset. The public contract is the observation semantics: Taxim produces RGB in the declared channel order, FOTS produces marker flow in the same task frame, both streams share the policy timestamp, and the policy consumes the history length H specified in Table 8. The release manifest should pin the configuration path and commit hash used for each reported run.

D.3. Calibration Checks and Known Limits

The pad collision thickness is tuned per USD asset by replaying real trajectories and checking contact onset, patch area, persistence, and Taxim/FOTS activation; Table 15 lists these checks and the quantities to report. The collision mesh supplies physics; the depth image supplies tactile rendering. The exact commercial sensor construction is unavailable, so camera pose and clipping values are alignment parameters for the rendered observation and operating range rather than reported hardware specifications.

Table 15: Replay checks for the tactile observation interface. Fill the status column from the frozen calibration report.

Check	Status	Reported quantity
Contact onset	*TBD*	onset error (ms or steps)
Contact-patch persistence	*TBD*	overlap / duration
Force scale	*TBD*	mean and maximum error (N)
Marker response	*TBD*	displacement correlation
Channel and timestamp alignment	*TBD*	RGB/flow ordering and lag

E. Reconstruction and Scene Verification

The workspace appearance is represented by 3D Gaussians, while explicit meshes provide object collision. A single registration transform aligns the scanned appearance with the robot base; the Gaussian tabletop is kept below the collision tabletop to avoid visual interpenetration. Gaussian assets never participate in contact.

During cleanup, the scene is checked with static-stage assertions, a headless task initialization, an offline camera probe, and the fixed-view/orbit-view renders planned in Fig. 11. If the real and collision tabletops differ in size, base alignment takes priority and the edge mismatch is retained. Known approximations include wide-angle Gaussian haze, small registration margins, and the absence of dynamic lighting on the Gaussian layer.

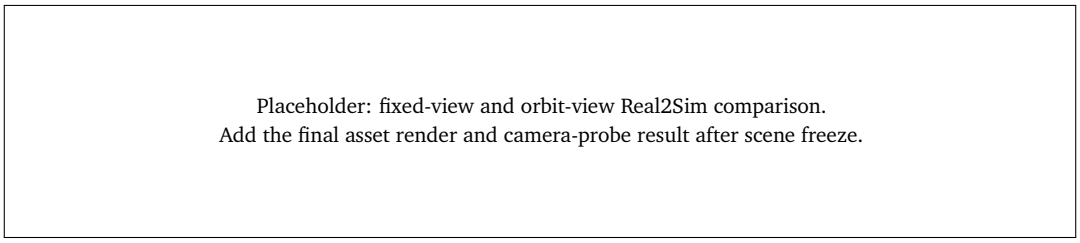


Figure 11: Planned visual verification of the reconstructed workspace.

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